CHEMISTRY GRADE 10: INTRODUCTION TO CHEMISTRY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 1.0: Inorganic Chemistry
Sub-Strand	Sub-Strand 1.1: Introduction to Chemistry
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Definition of Chemistry as a field of science – Branches of Chemistry (Organic, Inorganic, Physical, Analytical, Biochemistry, Nuclear, Industrial/Applied Chemistry) – Careers in Chemistry and how gender stereotyping influences career choices – Chemistry in our daily lives — agriculture, pharmaceutical industry, medicine, manufacturing industry, food industry, nuclear chemistry, entertainment industry, sports industry – Meaning of drug, prescription, dosage and substance use – Effects of drug and substance use in day-to-day life – Learner rights and responsibilities to a safe and healthy learning environment – Consumer protection — labels on manufactured products (marks of quality, expiry dates, nutritional information, preservatives, messages of caution)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the meaning of Chemistry as a field of science, b) explore the role of Chemistry in day-to-day life, c) examine the effects of drug and substance use in day-to-day life, d) advocate for a safe and healthy learning environment, e) identify career opportunities related to Chemistry and critically evaluate how gender stereotyping influences career choices.
Core Competencies	– Communication and Collaboration: The learner explores teamwork with peers by contributing to group decision making through recognising and acknowledging the value of others' ideas during discussions — demonstrated during peer discussions on the meaning of Chemistry, its branches, and the importance of Chemistry in daily Kenyan contexts (e.g., tea farming in Kericho, fish processing at Lake Victoria). – Self-efficacy: The learner effectively communicates while presenting the findings on career opportunities related to Chemistry and how gender stereotypes influence career choices in plenary — demonstrated when individual learners present research findings on careers such as pharmacist, food scientist, or agricultural chemist to the whole class. – Digital Literacy: The learner uses digital technology effectively to search for information on various career opportunities related to Chemistry — demonstrated when learners use the ARES Education platform and other digital resources to research Chemistry careers and compile findings. – Citizenship: The learner demonstrates critical and constructive dialogue while brainstorming with peers on the consumer rights

	to information on drug prescription, substance use and knowledge on their side effects for awareness — demonstrated during the drug and substance use discussions and the consumer-protection label-reading activity.
Core Values	– Respect: The learner shows acceptance while appreciating the opinion of each member as they discuss in groups the various branches of Chemistry — evident when learners take turns sharing ideas without interrupting peers and acknowledge contributions from all group members regardless of gender or background.
Science & Engineering Practices	– SEP 1 — Asking Questions and Defining Problems: Learners generate and refine questions about what Chemistry is and why it matters, anchored to the puzzling observation that everyday Kenyan products (ugali flour, fertiliser, medicine) are all shaped by Chemistry. – SEP 2 — Developing and Using Models: Learners create and revise a class concept-map model showing how the branches of Chemistry connect to real-world Kenyan industries and careers throughout the unit. – SEP 6 — Constructing Explanations and Designing Solutions: Learners construct explanations for why matter changes state (using the PhET simulation) and connect those explanations to the definition and scope of Chemistry. – SEP 7 — Engaging in Argument from Evidence: Learners use evidence from product labels and drug-information readings to argue for consumer rights and responsible substance use, supporting their claims with specific data (expiry dates, dosage instructions, nutritional information). – SEP 8 — Obtaining, Evaluating, and Communicating Information: Learners read, evaluate, and communicate information from print and electronic media about Chemistry careers, drug effects, and the importance of Chemistry — culminating in a poster presentation to peers or the school community.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Safety and Security): Learners brainstorm their rights and responsibilities to a safe and healthy learning environment, connecting Chemistry knowledge to personal safety decisions such as avoiding misuse of household chemicals and prescription drugs. – Health Promotion Issues (Drug and Substance Use): The learner discusses with peers the meaning of drug prescription, dosage, substance use and their effects in day-to-day life — contextualised through the Kenyan reality of readily available over-the-counter medicines and the risks of tobacco, alcohol, and illicit drug use among youth. – Socio-Economic and Environmental Issues (Consumer Protection): The learner discusses with peers the information on samples of containers of manufactured products and explores the labels on them for marks of quality, expiry dates, nutritional information, preservatives and messages of caution for consumer protection awareness — using locally available Kenyan products such as Delmonte juice, Unga maize flour, and Panadol packaging.
Career Connections	– Pharmacist: Sub-strand connects through the study of drugs, dosage, prescriptions, and the pharmaceutical industry — learners explore how pharmacists use Chemistry to prepare and dispense medicines across Kenya's health system. – Agricultural Chemist / Agronomist: Sub-strand connects through the importance of Chemistry in agriculture — learners examine how fertilisers, pesticides, and soil chemistry support food production in the Rift Valley and other farming regions. – Food Scientist / Nutritionist: Sub-strand connects through Chemistry in the food industry — learners analyse product labels on Kenyan food items (e.g., Brookside dairy products, Pembe flour) examining preservatives, nutritional content, and quality marks. – Chemical Engineer / Industrial Chemist: Sub-strand connects through Chemistry in manufacturing — learners explore how industries such as East African Breweries, KPLC, and Bamburi Cement apply Chemistry in production processes. – Medical Laboratory Scientist / Doctor: Sub-strand connects through Chemistry in medicine — learners discuss how blood tests, diagnostic reagents, and drug interactions rely on chemical knowledge. – Environmental Chemist: Sub-strand connects through the role of Chemistry in monitoring and protecting the environment — relevant to water quality at Lake Victoria and air quality in Nairobi's industrial areas. – Sports Scientist / Nutritionist: Sub-strand connects through Chemistry in the sports industry — learners discuss how Kenyan athletes such as marathon runners use sports nutrition and performance supplements rooted in biochemistry.

Focus for Lessons	This unit introduces Grade 10 learners to Chemistry as a field of science by anchoring learning in the puzzling observation that ordinary matter — the water in Lake Victoria, the ugali on a Kenyan family's table, the medicine in a clinic — all behave and change in ways that Chemistry explains. Over 10 lessons, learners move from defining Chemistry and mapping its branches, to investigating its role in Kenyan daily life, careers, and consumer protection, and finally to critically examining drug and substance use. The unit uses a phenomenon-driven, collaborative inquiry approach, with learners building and revising a shared concept-map model throughout, culminating in a community-awareness poster project on drug and substance use.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?" KICD KEY INQUIRY QUESTIONS: 1. What is Chemistry and why is it important in our daily lives? 2. How does drug and substance use affect our health and well-being?
Anchoring Phenomenon	ANCHORING PHENOMENON — "Why does the chapati dough on the jiko change into a firm, golden disc — and can we ever get the dough back?" At the start of Lesson 1, the teacher displays (or plays a short video of) a ball of raw chapati dough being placed on a hot jiko in a Kenyan kitchen. Students observe that the soft, pliable dough transforms into a firm, browned, aromatic chapati. The teacher then asks: "Can we reverse this? Can we turn the chapati back into dough?" Students quickly recognise that we cannot — something irreversible has happened. This is puzzling because the ingredients (wheat flour, water, oil, salt) are still physically present, yet the product is fundamentally different. The phenomenon is deliberately simple and universally familiar to Kenyan learners, yet it immediately raises deep questions: What exactly changed? Was it the matter itself or the energy? What field of science studies this? This single observable event anchors the entire unit — the definition of Chemistry as the study of matter and its changes, the branches that explain different aspects of those changes, the industries (food, pharmaceutical, agricultural) that harness such changes, and ultimately the careful, science-based approach needed when introducing chemical substances (drugs) into the human body.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Defining Chemistry: Learners observe the chapati-on-jiko phenomenon and the PhET States of Matter simulation. They record initial observations and puzzling questions on sticky notes for the Driving Question Board (DQB). Through peer discussion guided by the teacher, learners arrive at a working definition of Chemistry as the study of matter and its changes, grounded in what they observed. Lesson 2 — Branches of Chemistry: Learners map the branches of Chemistry (Organic, Inorganic, Physical, Analytical, Biochemistry, Nuclear, Industrial) onto real Kenyan contexts — e.g., Biochemistry explains why chapati browns (Maillard reaction, introduced at conceptual level); Industrial Chemistry explains how Bamburi Cement is made. Learners revise and expand the class concept-map model to include branches. Lesson 3 — Chemistry in Agriculture and Food: Learners brainstorm and research (using ARES platform resources) how Chemistry underpins Kenyan agriculture (fertilisers in the Rift Valley, pesticides, soil pH) and the food industry (preservatives in Pembe flour, fermentation in uji). They add "Importance of Chemistry — Agriculture & Food" to the DQB and concept map. Lesson 4 — Chemistry in Medicine, Pharmaceuticals and Manufacturing: Learners investigate how Chemistry is applied in medicine (blood tests, anaesthetics), the pharmaceutical industry (drug synthesis, KEBS standards), and manufacturing (EABL, Bidco). A reading on the importance of Chemistry in these sectors is used as the evidence source. Learners update the concept

	map and DQB. Lesson 5 — Chemistry in Other Industries (Nuclear, Sports, Entertainment, Energy): Learners research and discuss Chemistry's role in nuclear energy (context: Kenya's energy mix and KPLC), sports nutrition (Kenyan marathon runners, supplements), entertainment (dyes, special effects), and environmental chemistry. Learners present findings in a brief plenary. DQB and model updated. Lesson 6 — Careers in Chemistry and Gender Stereotyping: Learners use the ARES platform and print resources to research careers in Chemistry. They critically examine how gender stereotypes in Kenyan society influence who pursues Chemistry careers. Learners present findings and debate: "Should Chemistry careers be gendered?" Self-efficacy competency is the focus. DQB updated with career-related questions. Lesson 7 — Meaning of Drug, Prescription, Dosage and Substance Use: Learners read a science-of-drugs article and discuss key terms: drug, prescription, dosage, over-the-counter vs. prescription medicine, substance use vs. abuse. They examine a Panadol and a Metformin label for dosage and caution information. Citizenship competency is the focus. DQB updated. Lesson 8 — Effects of Drug and Substance Use (Tobacco Focus): Learners watch the video on the effects of tobacco use and analyse the chemical mechanisms at a conceptual level (tar, nicotine, carbon monoxide). They argue from evidence: "The tobacco plant is natural — so why is smoking harmful?" Learners connect this back to the anchoring phenomenon: Chemistry explains why natural substances can cause irreversible changes in the body, just as heat causes irreversible changes in chapati dough. Lesson 9 — Consumer Protection and Safe Learning Environment: Learners examine labels on 3–4 Kenyan manufactured products (food, cleaning agent, medicine) for KEBS quality marks, expiry dates, nutritional information, preservatives, and safety warnings. They brainstorm their rights and responsibilities to a safe and healthy learning environment, linking Chemistry knowledge to personal safety (avoiding misuse of lab chemicals, household chemicals, medications). Lesson 10 — Project: Poster on Drug and Substance Use & Unit Synthesis: Learners develop posters to sensitise peers and the community on the risks of drug and substance use, using evidence gathered across the unit. The class also conducts a final review of the DQB — crossing off questions answered and noting what remains for future study. The concept-map model is finalised and displayed. Learners reflect on how the chapati phenomenon connects to everything they have learned about Chemistry and its role in their lives.
Supporting Phenomena	– Supporting Phenomenon 1 — PhET States of Matter Simulation: Learners manipulate the PhET "States of Matter: Basics" simulation to observe water molecules transitioning between solid (ice on Mt. Kenya), liquid (tap water), and gas (steam from a sufuria). This supports the anchoring phenomenon by showing that matter changes are driven by energy changes at the particle level — laying the conceptual foundation for defining Chemistry. – Supporting Phenomenon 2 — Video: "What is Chemistry?": A short curated video showing Kenyan scientists, pharmacists, farmers, and engineers at work, each explaining how Chemistry underpins their daily professional decisions. This makes the definition of Chemistry concrete and locally relevant. – Supporting Phenomenon 3 — Product Label Investigation: Learners examine physical labels from common Kenyan manufactured products (e.g., a packet of Pembe maize flour, a Panadol blister pack, a bottle of Dettol) and notice the abundance of chemical information — ingredients, preservatives, dosage warnings, expiry dates, KEBS quality marks. The puzzle: "Why does every product in our home carry chemical information — and why does it matter if we ignore it?" This supports the consumer-protection and drug-use strands.

	– Supporting Phenomenon 4 — Video: Effects of Tobacco Use: A video showing the chemical effects of tobacco smoke on lung tissue, framed around the question: "The tobacco plant is natural — so why is smoking harmful?" This directly supports the drug and substance use lessons and connects to the unit's advocacy outcome.

LESSON 1 (80 minutes): Anchoring Phenomenon & Defining Chemistry: What Happened to the Chapati Dough?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit anchoring phenomenon and establish a working definition of Chemistry as the study of matter and its changes.
Knowledge	Learners explain the meaning of Chemistry as a field of science (KICD SLO a), connecting the definition to observable, everyday changes in matter such as the transformation of chapati dough on a jiko.
Skills	Learners record structured observations, generate questions on sticky notes for the Driving Question Board, and discuss with peers the meaning of Chemistry as a field of science (KICD Suggested Learning Experience 1).
Attitudes	Learners demonstrate curiosity and respect by acknowledging the value of peers' ideas during group discussions (KICD Value — Respect; Core Competency — Communication and Collaboration).
Key Inquiry Question	What is Chemistry and why is it important in our daily lives? (KICD Key Inquiry Question 1)
Purpose in Storyline	Lesson 1 launches the anchoring phenomenon — chapati dough transforming irreversibly on a jiko — and opens the Driving Question Board. Learners produce their first attempt at defining Chemistry from evidence, a definition they will refine across all 10 lessons of Sub-Strand 1.1.
Safety Notes	If a live jiko demonstration is used, ensure adequate ventilation and a 1-metre learner exclusion zone around the heat source. No learner should handle the jiko or cooking surface. If using a video instead, no physical safety precautions are required beyond standard screen-viewing guidelines. Remind learners that the PCIs on safety and security apply inside the classroom too — everyone has a right to a safe learning environment (KICD PCI — Life Skills: Safety and Security).

B. LESSON OVERVIEW
Learners observe (via video or live demonstration) a ball of raw chapati dough placed on a hot jiko transforming into a firm, golden, aromatic chapati. They immediately recognise the change is irreversible and record puzzling questions. Through peer discussion and a PhET States of Matter simulation, they gather evidence about what matter is and how it changes, arriving collaboratively at a working definition of Chemistry. The lesson closes by opening the class Driving Question Board (DQB) with learners' own sticky-note questions, anchoring the entire unit storyline.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Anatomy of a skeletal muscle cell] Source: Khan Academy (English	Before the phenomenon is revealed, each learner receives a half-sheet of paper. The teacher places a covered bowl on the demonstration table and asks: 'Inside this bowl is something you have seen in a Kenyan kitchen	1. Distribute prediction half-sheets before any reveal. Say exactly: 'Do not call out answers yet — I want every brain working independently first. You have 3 minutes. Go.' Set a visible timer. 2. After 3 minutes, say: 'Turn and tell your shoulder	Think-Write-Pair-Share (TWPS): Learners first commit ideas to paper individually (reducing anchoring bias from dominant voices), then refine through partner dialogue, then hear class-level variation. This surfaces prior	Collect prediction sheets at the end of Phase 1. Observable indicator: Can the learner identify at least one property of matter (e.g., shape, texture, smell, colour) that might change during cooking? Learners who write only 'it gets

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	hundreds of times. Before I show you, I want you to think: what kinds of changes happen to food when we cook it? Write or draw TWO things you think could change and ONE thing you think stays the same.' Learners write silently for 3 minutes, then share with a shoulder partner. Selected pairs share with the class. Learners hold their prediction sheets — they will revisit them at the end of the lesson to see whether their ideas were confirmed or challenged.	partner ONE thing you wrote.' Allow 60 seconds of partner talk. 3. Cold-call 4 pairs (not volunteers — use the class register to select): 'Pair at the back-left, what did you predict stays the same?' Probe with: 'Why do you think that?' Allow 10-second WAIT TIME after each response before moving on. 4. Do NOT confirm or deny predictions. Say: 'Hold those ideas — our phenomenon today will give you evidence to test them.' 5. Record 3–4 student prediction phrases on the board under the heading 'Our Predictions' — these stay visible all lesson.	conceptions about matter and change that the teacher uses to calibrate instruction.	hot' need scaffolding in Phase 3 to distinguish physical properties from chemical identity.
Observe Phase 📹 **VIDEO:** Anatomy of a skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Chapati Phenomenon (20 min): The teacher uncovers the bowl to reveal a ball of raw chapati dough, then plays a 2-minute video clip (or conducts a live demonstration) of the dough being placed on a hot jiko and cooking into a chapati. Learners use the 'See–Think–Wonder' observation frame printed on their worksheet: they record (i) what they SEE changing, (ii) what they THINK is causing the change, and (iii) what they WONDER — questions they cannot yet answer. After the video, the teacher asks: 'Can we reverse this? Can we turn the chapati back into dough?' Learners discuss in groups of 4 for 2 minutes, then the class votes by show of hands: Yes / No / Not sure. PART B — PhET States of Matter Simulation (15 min): Learners open the PhET 'States of Matter: Basics' simulation (pre-loaded on ARES offline devices). Working in pairs, they select 'Water', then move the temperature slider from Solid → Liquid → Gas and back. They	1. Before playing the video, say: 'You are scientists now. Scientists observe carefully before they explain. Your only job for the next 2 minutes is to watch and write — no talking.' Play the clip. 2. After the clip, give 2 minutes of silent See-Think-Wonder writing. Then say: 'Share your WONDER column with your group. Choose your group's most puzzling question.' Allow 3 minutes of group discussion. 3. For the reversal vote, say: 'Hands up — who says YES, we can reverse it?' Count aloud. 'Hands up for NO.' Count aloud. Record numbers on the board. Then cold-call one learner from each camp: 'Achieng, you voted NO — what is your evidence from the video?' WAIT 12 seconds. 4. Transition to PhET: say 'Now let us zoom in — zoom in so small we can see individual particles of matter. Open the simulation.' Circulate and ask probing questions: 'What happens to the particles when you heat them? Does this remind you of anything in the chapati video?' 5. With 3	See–Think–Wonder (STW) + Particle-Level Observation: STW externalises the difference between raw observation (See), inference (Think), and genuine scientific questions (Wonder). Pairing it with the PhET simulation bridges the macroscopic phenomenon (chapati changing) to the particle-level model that defines Chemistry — matter is made of particles whose arrangement and interactions explain observable change. This dual-scale observation is the conceptual engine of the entire unit.	Collect or photograph the See-Think-Wonder worksheets. Observable indicators: (i) 'See' column should list specific sensory observations (colour, texture, aroma, rigidity) — not vague terms like 'it cooked'. (ii) 'Wonder' column should contain at least one genuine question (e.g., 'Why can't we un-cook it?', 'What happened inside the flour?'). (iii) PhET particle diagrams should correctly show particles closer together in solid than gas. Flag learners whose diagrams show no difference between states — they need targeted follow-up in Lesson 2.

	record on their worksheet: how particle arrangement changes, whether each change is reversible, and how this compares to the chapati. Learners annotate a particle diagram for solid, liquid, and gas states.	minutes remaining in Part B, call the class back: 'Freeze your simulations. Look at your particle diagrams. In one sentence, tell your partner: what is matter doing as it changes state?' Cold-call 3 pairs. WAIT 10 seconds each.		
Explain Phase 📹 **VIDEO:** Anatomy of a skeletal muscle cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Peer Discussion: What field of science studies this? (10 min): Each group of 4 receives a set of 6 prompt cards showing images and one-line descriptions: a pharmacist dispensing medicine in Nairobi, a farmer in Kericho testing soil, a food technologist at a uji processing plant in Eldoret, a materials engineer at a cement factory in Athi River, a nuclear scientist, and a sports nutritionist advising a Kenyan marathon runner. Groups discuss: 'Which of these people needs to understand how matter changes? What would we call the science they all share?' Groups write a draft definition of Chemistry in one sentence on a mini-whiteboard or large sticky note. PART B — Whole-Class Definition Synthesis (8 min): Each group shares their draft definition. The teacher writes key recurring words on the board (matter, change, energy, substances, properties). Guided by the teacher, the class negotiates a single working definition. The teacher then reveals the KICD-aligned definition: 'Chemistry is the study of matter, its properties, composition, and the changes it undergoes.' Learners write this in their notebooks. PART C — Connecting back to the phenomenon (5 min): Learners answer in writing: 'Using our working definition of Chemistry, write two sentences explaining	1. Distribute prompt cards. Say: 'You have 5 minutes in your groups. Your task: find the common thread — what science connects all these people and jobs? Write your answer as a complete sentence.' Set timer. 2. After 5 minutes, use a round-robin call (not volunteers): call each group's nominated speaker in turn. After each definition, ask the class: 'What word did they use that we should keep? Thumbs up if you agree with including it.' Build the board list of key words collaboratively. 3. Once the board list is ready, say: 'Now let us write THE definition together. I will start: Chemistry is the study of…' Pause for 10 seconds. Accept completions from the floor. Guide toward the KICD wording without simply dictating it. 4. After revealing the KICD definition, say: 'Compare it to what your group wrote. Circle any word in the official definition that your group did NOT include. That word is your learning edge today.' 5. For Part C, say: 'This is individual thinking time — no copying from your neighbour. You have 4 minutes.' Circulate and read over shoulders. If a learner is stuck, prompt: 'Look at the word 'changes' in the definition. What changed about the chapati?' WAIT 10 seconds before offering further help.	Collaborative Definition Building (Concept Negotiation): Rather than receiving the definition passively, learners construct it from evidence (prompt cards, phenomenon, simulation data) and then compare their version to the authoritative KICD definition. The 'learning edge' exercise (circling missing words) makes the gap between naïve and scientific understanding explicit and personal — a metacognitive move that deepens retention. Connecting the definition back to the chapati phenomenon ensures Chemistry is never abstract: it is always the science that explains something real.	Read the Part C written responses. Observable indicator: A proficient response names at least two of the four elements of the KICD definition (matter / properties / composition / changes) and correctly applies them to the chapati transformation (e.g., 'Chemistry studies how matter changes — the dough changed its properties when heated, so Chemistry explains why the chapati is now firm and cannot return to dough'). Learners who write only 'because Chemistry is about cooking' need a follow-up probe question in the next lesson.

	why Chemistry — and not another science — is the best tool for explaining what happened to the chapati dough on the jiko.'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Anatomy of a skeletal muscle cell] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes of different colours — one colour for 'Questions about the chapati / matter changes' and another colour for 'Questions about Chemistry in my life or career'. Learners write one question per sticky note (minimum one of each colour). They then walk up to the class DQB — a large sheet of paper or whiteboard section divided into columns matching future lesson topics (Definition, Branches, Careers, Daily Life, Drug & Substance Use) — and place each sticky note in the column they think it belongs to. The class then does a 'gallery walk' of 3 minutes, reading peers' questions silently and placing a dot sticker on any question that also puzzles them. The teacher highlights 3–4 questions that will be directly addressed in upcoming lessons, creating explicit anticipation. The Driving Question 'Why does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices?' is written in large text at the top of the DQB and stays on the wall for all 10 lessons.	1. Distribute sticky notes and explain the colour code clearly. Say: 'On your YELLOW note — one question about what happened to the chapati or about matter. On your BLUE note — one question about Chemistry in your life, your future career, or how it affects your health.' Allow 3 minutes of silent individual writing. 2. Invite learners to place notes on the DQB. If a learner is unsure which column, say: 'Take your best guess — we can move it later as we learn more. Scientists reorganise their thinking all the time.' 3. Conduct the gallery walk. Say: 'Walk quietly, read, and dot-sticker any question that is also YOUR question. You have 3 minutes.' 4. After the gallery walk, read aloud the top 3 most-dotted questions. Say: 'These are the questions our class most wants to answer. Let us see if we can answer them by the end of Sub-Strand 1.1.' Write those 3 questions in a special box on the DQB labelled 'Our Top Questions'. 5. Point to the Driving Question at the top and read it aloud together as a class. Say: 'Every lesson this term, we will come back to this board and add what we have learned. By Lesson 10, this board will be full of your evidence.'	Driving Question Board (DQB) Launch: The DQB is a public, cumulative record of the class's evolving understanding. Launching it in Lesson 1 — before any instruction — communicates to learners that their questions are the engine of the unit, not the teacher's agenda. The dot-sticker voting creates social proof that curiosity is shared, reducing the anxiety of 'I don't know.' Categorising questions by future lesson topics gives learners a cognitive map of the sub-strand and creates lesson-to-lesson continuity.	Photograph the DQB at the end of Lesson 1. Observable indicators: (i) Every learner has contributed at least one yellow and one blue sticky note (participation check). (ii) Yellow notes should reference observable phenomena (change, heat, smell, irreversibility) — not just restate the lesson topic. (iii) Blue notes should show some awareness that Chemistry connects to life beyond school (food, medicine, farming, jobs). Questions that are very vague ('What is chemistry?') indicate learners who need the phenomenon revisited at the start of Lesson 2.
Model Building Phase 📹 **VIDEO:** [Anatomy of a skeletal muscle	Each learner opens to a blank page in their Science Notebook labelled 'My Chemistry Model — Lesson 1'. They draw and annotate a two-panel model: Panel 1 shows the chapati dough	1. Say: 'Scientists do not just write definitions — they build models. A model is your best current explanation, drawn out. It does not have to be perfect. It has to be honest about what you know AND	Cumulative Two-Panel Model (Macro + Particle Level): Beginning the model in Lesson 1 — before learners have complete knowledge — is intentional. It externalises current thinking,	Review photographed models against three indicators: (i) Both panels present (BEFORE and AFTER) — learners who draw only one panel need a model-building scaffold in Lesson 2. (ii) Particle-

cell Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	BEFORE heating (macroscopic: a soft ball; particle-level: their best guess at how the particles are arranged). Panel 2 shows the chapati AFTER heating (macroscopic: a firm, golden disc; particle-level: their best guess at how the particles are now arranged differently). Below the panels, learners write: (i) 'What I am sure of:' (ii) 'What I am still unsure of:' (iii) 'What Chemistry concept might explain this:' Learners share their models with their group for 2 minutes of peer feedback using the sentence stem: 'I notice your model shows ___. I wonder if you considered ___.' The teacher collects or photographs models. These SAME models will be revised in every subsequent lesson as understanding deepens — by Lesson 10, learners will have a sophisticated particle-level explanation of the chapati change.	what you do not know yet.' 2. Display a simple two-panel model template on the board (hand-drawn is fine). Label it explicitly: 'BEFORE' and 'AFTER'. Say: 'Draw both levels — what you can see with your eyes (macroscopic) and what you imagine the tiny particles are doing (particle level). Use the PhET simulation to help you with the particle level.' Allow 8 minutes of silent individual drawing. 3. During drawing, circulate and use targeted prompts: 'How tightly packed are the particles in the raw dough compared to the cooked chapati? What does the PhET simulation suggest?' WAIT 10 seconds after each prompt. Do not draw for learners. 4. After 8 minutes, say: 'Share your model with your group. Use this sentence: I notice your model shows ___, I wonder if you considered ___.' Allow 2 minutes. 5. Collect or photograph all models. Say: 'I am not grading these today. I am saving them so YOU can see how much your thinking grows. We will add to these models in every lesson.' 6. Close the lesson by returning to the Driving Question: 'Say the driving question with me.' Read it aloud together. 'Today we took the first step. What is Chemistry? We said it is the study of matter and its changes. The chapati on the jiko is our proof that matter changes — and Chemistry is the science that explains HOW and WHY.'	making misconceptions visible and addressable. The 'unsure of' section normalises scientific uncertainty and sets up genuine motivation to learn. Revisiting and revising the same model across 10 lessons builds a powerful metacognitive record of conceptual growth, directly aligned with the NGSS Science and Engineering Practice of Developing and Using Models.	level attempt present — even a rough attempt (circles drawn close together vs. spread apart) shows engagement with the simulation; blank particle sections indicate the learner did not connect the PhET activity to the phenomenon. (iii) 'Still unsure of' section completed honestly — this is the highest-value formative data: questions like 'I don't know why the particles can't go back' signal readiness for the Lesson 2 discussion on reversible vs. irreversible changes.

D. TEACHER REFLECTION

After Lesson 1, consider the following reflective questions: (1) Did every learner contribute a sticky note to the DQB, or did some learners remain passive? If passive learners were visible, plan a brief one-on-one check-in at the start of Lesson 2 using their See-Think-Wonder sheet as a conversation anchor. (2) Review the

photographed models: how many learners attempted a particle-level diagram? If fewer than 70% attempted the particle level, consider spending 5 additional minutes at the start of Lesson 2 revisiting the PhET simulation as a whole class before continuing. (3) Which 'Wonder' questions from learners were genuinely surprising or showed prior knowledge you did not expect? Note these and weave them explicitly into Lessons 2–4. (4) Did the class successfully arrive at a definition close to the KICD wording through discussion, or did you find yourself dictating it? If the latter, consider providing more varied prompt card images in future — include more Kenyan local contexts (e.g., a tea processor in Kericho, a salt harvester at Lake Magadi). (5) Note any gender patterns in participation: did female learners engage equally during cold-calling and model sharing? If not, deliberately sequence cold-calls to amplify female voices in Lesson 2, connecting to the KICD Core Competency on self-efficacy and the sub-strand's discussion of gender stereotyping in Chemistry careers.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Raw chapati dough placed on a hot jiko transforms irreversibly into a firm, golden, aromatic chapati. In the PhET States of Matter simulation, water particles move from tightly packed (solid) to loosely spaced (gas) and back — showing that some matter changes are reversible at the particle level, while the chapati change is not.
What did I learn?	Chemistry is the study of matter, its properties, composition, and the changes it undergoes. Matter is anything that has mass and takes up space. Studying Chemistry helps us understand why and how matter changes — from chapati on a jiko to medicines in our bodies. Some changes in matter are reversible; others (like cooking dough) are not.
How does this explain the phenomenon?	Chemistry — and not Biology or Physics alone — is the best tool for explaining the chapati transformation because the change involved the properties, composition, and irreversible chemical change of the matter (dough) itself. Chemistry helps us understand how drugs interact with the body at a molecular level, and why understanding matter changes leads to safer, smarter choices in daily life.

LESSON 2 (80 minutes): Branches of Chemistry: Mapping the Science to Our World	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explore the branches of Chemistry and connect each branch to real Kenyan contexts, industries, and daily-life phenomena — including the chapati browning phenomenon.
Knowledge	Learners will be able to: (a) identify and describe the main branches of Chemistry (Organic, Inorganic, Physical, Analytical, Biochemistry, Nuclear, Industrial); (b) explain how each branch relates to specific real-world applications in Kenya; (c) explain at a conceptual level how Biochemistry accounts for the browning and irreversibility of chapati cooking (Maillard reaction).
Skills	Learners will be able to: (a) discuss with peers the branches of Chemistry; (b) map branches of Chemistry onto Kenyan industrial and daily-life contexts using a concept map; (c) use print or electronic media to identify career opportunities linked to specific branches.
Attitudes	Learners will: (a) show respect by accepting and acknowledging peers' ideas during group discussions (Value: Respect); (b) appreciate the relevance of Chemistry to their own lives, communities, and future careers in Kenya.
Key Inquiry Question	What is Chemistry and why is it important in our daily lives?
Purpose in Storyline	In Lesson 1, learners established that Chemistry is the study of matter and its changes, and they anchored that definition to the chapati-on-the-jiko phenomenon. In Lesson 2, learners discover that Chemistry is not a single discipline — it is a family of specialised branches. Each branch offers a different lens through which to explain the same phenomenon and the wider world. By the end of this lesson, learners understand WHICH branch of Chemistry explains WHY chapati cannot be un-cooked, building explanatory power toward answering the driving question. The concept-map model built in Lesson 1 is revised and expanded to include all branches.
Safety Notes	No hazardous materials are used in this lesson. If a hot jiko or stove is referenced as a demonstration prop, ensure it is placed on a stable, heat-resistant surface away from learners. No open flames during the main activity. Remind learners that even familiar substances (cooking oil, baking soda) are chemical agents and should be handled respectfully.

B. LESSON OVERVIEW
Learners begin by revisiting the class Driving Question Board from Lesson 1 and recalling the chapati phenomenon. Through a structured Predict–Observe–Explain cycle, they first predict which area of science would explain the chapati's browning and irreversibility, then gather evidence by reading a set of short Kenyan-context cards that introduce each branch of Chemistry (Organic, Inorganic, Physical, Analytical, Biochemistry, Nuclear, Industrial). In the Explain phase, groups map each branch onto a large concept map and connect Biochemistry explicitly to the chapati phenomenon via a conceptual introduction to the Maillard reaction. Learners update the Driving Question Board and conclude by revising the class model to show how the branches of Chemistry together explain changes in matter around us every day.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Biological redox]()	Learners are shown a projected split image: on the left, raw chapati dough on a floured board in a	Display the split chapati image and say: 'Last lesson we agreed Chemistry is the study of matter	Think-Pair-Share with DQB Prediction Posting. This strategy activates prior knowledge and	Observable indicator: Every learner has a written prediction on paper before sharing. Teacher

[reactions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Nairobi kitchen; on the right, a golden-brown chapati resting on a jiko grill. The teacher re-reads the anchoring question from the DQB: 'Can we ever get the dough back?' Each learner silently writes a response to the prompt: 'Which branch or area of science do you think explains WHY the dough changed colour, hardened, and became impossible to reverse? Write the name of the branch if you know it — or describe what that science would need to study.' Learners then share with a shoulder partner (Think-Pair-Share). Selected pairs report to the class. Predictions are recorded on sticky notes and posted on a dedicated 'Our Predictions' column on the DQB.	and its changes. Today we go deeper. Look at these two images.' Point to each side deliberately. Ask: 'Before I tell you anything new — which BRANCH of science, or which TYPE of Chemistry, do you think would explain this transformation? Take 60 seconds to write your answer silently.' [WAIT TIME: 60 seconds — no prompting.] Then say: 'Share with your shoulder partner for 30 seconds each.' [WAIT TIME: 60 seconds for pairs.] Cold-call 4 pairs (vary gender and seating zone): 'Pair at the front-left — what did you predict and WHY?' After each response, press with: 'Interesting — what evidence from the image makes you say that?' Do NOT confirm or correct predictions yet. Record all prediction keywords (e.g., 'Biology,' 'Cooking science,' 'Biochemistry,' 'Physics of heat') on the DQB sticky-note column. Close by saying: 'Hold your predictions — let's gather some evidence.'	creates cognitive dissonance — learners who say 'Biology' or 'Physics' will need to revise their mental models when they encounter the branches of Chemistry. Posting predictions publicly creates accountability and a reference point for revision at the end of the lesson.	scans written responses during pair-share to identify: (a) learners who already know 'Biochemistry' (extend — ask them to explain the reaction), (b) learners who conflate Chemistry with Biology or Physics (target — use their prediction as a bridge during Explain phase). Note at least 2 misconceptions heard to address explicitly in Phase 3.
Observe Phase 📹 **VIDEO:** [Biological redox reactions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of 4–5. Each group receives a set of 7 colour-coded Branch Cards (one per branch), each printed with: (1) the branch name, (2) a one-paragraph plain-language description of what that branch studies, and (3) a specific Kenyan example. The cards are: ORGANIC CHEMISTRY — studies carbon-containing compounds; Kenyan example: the sugars and starch in maize grain that ferment to make busaa or are processed at Mumias Sugar Company. INORGANIC CHEMISTRY — studies non-carbon compounds and minerals; Kenyan example: the minerals in	Distribute Branch Card sets face-down. Say: 'You each have 7 cards. Do NOT flip them until I say go. When I say go, your group has 12 minutes to read every card, discuss it, and fill in your Branch Evidence Table. The last column — connection to chapati — might be blank for most branches, and that is perfectly fine. Go.' [Set a visible 12-minute timer.] Circulate continuously. At the 4-minute mark, crouch next to a group and ask quietly: 'Which branch surprised you most? Why?' At the 8-minute mark, prompt any group still blank on the last column: 'Read the Biochemistry card again — look at the words protein, sugar,	Jigsaw-style Branch Card Reading with Evidence Table. Reading authentic, Kenya-contextualised descriptions before formal instruction allows learners to build initial schema from evidence rather than passive note-taking. The Evidence Table structure (especially the 'Connection to Chapati' column) keeps the anchoring phenomenon live and ensures all new information is processed relative to a shared reference point.	Observable indicator: Completed Branch Evidence Tables with at least 6 of 7 branches filled in correctly. Teacher checks specifically that the Biochemistry row mentions 'proteins,' 'sugars,' 'heat,' and 'colour change' or equivalent phrases. Learners who leave the Biochemistry-chapati connection blank are flagged for targeted support in Phase 3. Teacher also listens for groups that conflate Organic Chemistry with Biology — a predictable misconception to address in the Explain phase.

	soil at Rift Valley farms that make fertiliser effective; how Bamburi Cement processes limestone. PHYSICAL CHEMISTRY — studies energy and rates of chemical reactions; Kenyan example: why ugali cooks faster on a gas jiko than on a wood fire (energy and reaction rates). ANALYTICAL CHEMISTRY — studies the composition and purity of substances; Kenyan example: how Kenya Bureau of Standards (KEBS) tests cooking oil for adulteration; how Nairobi Water Company tests drinking water. BIOCHEMISTRY — studies chemical processes in living organisms; Kenyan example: how the proteins and sugars in chapati dough react at high temperature to produce the golden-brown colour and new flavour compounds (Maillard reaction — the same process that browns nyama choma on a Maasai moran's grill). NUCLEAR CHEMISTRY — studies radioactive elements and nuclear reactions; Kenyan example: how radioactive iodine is used in thyroid cancer treatment at Kenyatta National Hospital. INDUSTRIAL CHEMISTRY — applies chemical processes at large scale for manufacturing; Kenyan example: how East African Breweries or Kenya Breweries Limited ferments and purifies products at industrial scale. Groups read all 7 cards, discuss, and complete a Branch Evidence Table in their exercise books with columns: Branch | What it Studies | Kenyan Example | Connection to Chapati Phenomenon (if any).	high temperature. Does that remind you of anything on the jiko?' [WAIT TIME: 15 seconds after this prompt before offering further help.] At the 12-minute mark, call time. Cold-call 3 different groups (not the same ones from Phase 1) to share one row each from their table: 'Group by the window — tell us about Industrial Chemistry and your Kenyan example.' Affirm accuracy, then ask the class: 'Does any other group have a different Kenyan example for Industrial Chemistry? Hands up.' Take 2 additional examples. Repeat for Biochemistry, asking specifically: 'What did the Biochemistry card say is happening when chapati turns brown?'		
Explain Phase 📹 **VIDEO:**	Part A — Class Concept Map Expansion (20 min): The teacher	For Part A, say: 'Our concept map from last lesson had one centre	Collaborative Concept Mapping + Analogical Reasoning (Maillard	Observable indicator: The completed class concept map has

Biological redox reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	projects or draws the Lesson 1 concept map on the board (centre node: 'Chemistry — the study of matter and its changes'). Learner volunteers come to the board one at a time to add each branch as a labelled node with a connecting line, writing the Kenyan example beside it. The class evaluates each addition — 'thumbs up / thumbs to the side' — and suggests corrections. Part B — The Maillard Reaction: Why Can't We Un-Cook the Chapati? (10 min): The teacher leads a whole-class explanation using a simple analogy and a visual diagram. The teacher draws on the board: 'Amino acids (from wheat protein) + Sugars (from wheat starch) + HEAT → NEW molecules (brown colour + aroma) + cannot go back.' The teacher explains: 'This is called the Maillard reaction — named after the French scientist Louis-Camille Maillard, but Kenyan food scientist researchers at the University of Nairobi study similar reactions in local foods every day. The new molecules formed are chemically different from the starting materials. There is no reverse reaction under normal conditions — which is why your mum cannot un-cook the chapati.' Learners annotate their concept maps with a special 'Chapati node' connected to Biochemistry, labelled: 'Maillard reaction — irreversible change.' Part C — Branches-to-Careers Bridge (5 min): Learners do a rapid-fire Think-Pair: 'For each branch, name ONE career a Kenyan young person could pursue.' Pairs share two examples with the class.	node. Today we are going to grow it. I need 7 volunteers — one for each branch.' Select volunteers ensuring gender balance. As each volunteer writes, ask the class: 'Is this branch correctly described? Thumbs up if yes, thumbs to the side if you want to suggest a change.' After all 7 nodes are added, ask: 'Which two branches are MOST closely connected to each other? Turn to your neighbour and decide — 10 seconds.' [WAIT TIME: 10 seconds.] Cold-call 3 pairs. Accept and discuss connections (e.g., Organic ↔ Biochemistry; Analytical ↔ Industrial). For Part B, draw the Maillard diagram slowly, narrating each step. Then ask: 'Now look at your prediction sticky note on the DQB. Who predicted Biochemistry? Raise your hand.' Acknowledge correct predictors warmly. Then ask: 'Who predicted Physics or Biology? That was logical thinking — heat IS involved (Physical Chemistry plays a role in the energy) and living organisms ARE involved (wheat is a plant). But the SPECIFIC explanation for the brown colour and irreversibility belongs to Biochemistry. Can you see why?' [WAIT TIME: 15 seconds.] Cold-call 2 learners who had incorrect predictions: 'Given what you now know, how would you revise your prediction?' For Part C, use rapid cold-call — point to 6 different learners quickly around the room for one career each. Write all on the board.	Diagram). The concept map expansion uses public knowledge construction — learners see their peers' thinking made visible and evaluate it, deepening engagement. The Maillard diagram uses a simple input → output arrow model (an analogy to a recipe) that is accessible before learners have studied chemical equations. Connecting back to the DQB prediction column gives learners an explicit opportunity to experience productive revision of ideas — a core scientific practice (NGSS SEP 6: Constructing Explanations).	7 branch nodes, each with a Kenyan example and at least one inter-branch connection arrow. Every learner's individual concept map includes the 'Chapati node' connected to Biochemistry. Teacher listens for correct use of the word 'irreversible' when describing the Maillard reaction. Exit check: Teacher asks 2 learners chosen at random to point to the concept map and say in one sentence: 'Biochemistry explains the chapati phenomenon because…' — learners must use the words 'protein,' 'sugar,' and 'irreversible' in their sentence.
Driving	Learners return to the class DQB.	Distribute two sticky notes	Driving Question Board —	Observable indicator: Every

Question Board (DQB) Creation 📹 **VIDEO:** Biological redox reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Working individually, each learner writes two items on separate sticky notes: (1) a GREEN 'Evidence' note — one piece of evidence from today's lesson that helps answer either Key Inquiry Question ('What is Chemistry and why is it important in our daily lives?' or 'How does drug and substance use affect our health and well-being?'); (2) an ORANGE 'New Question' note — one new question today's lesson has raised for them. Learners post both notes on the DQB in the correct columns. The teacher reads 3–4 notes aloud (a mix of evidence and questions) and briefly validates each, noting which new questions will be addressed in upcoming lessons.	(different colours if possible) to each learner. Say: 'You have 3 minutes. Write one piece of evidence on the green note — something from today that helps answer our driving question about Chemistry in daily life. Write one new question on the orange note — something today's lesson made you curious about.' [WAIT TIME: 3 minutes, silent individual writing.] Then say: 'Come post your notes on the board — green in the Evidence column, orange in the New Questions column.' Allow 2 minutes for posting. Read aloud 3 green notes, saying: 'This learner wrote: [read note]. This is strong evidence because it shows [connect to driving question].' Read 2 orange notes and say: 'Great question — we will come back to this in Lesson [X]. Keep watching for evidence.' Circle one orange note that asks about careers or drug interactions (likely present given the Branch Cards) and say: 'This question about [e.g., how Biochemistry is used in medicine] is going to be very important when we get to the topic of drugs and substance use later in this unit.'	Evidence and New Questions Protocol. This strategy (aligned with NGSS SEP 1: Asking Questions and Defining Problems, and SEP 7: Engaging in Argument from Evidence) makes learners' growing understanding publicly visible and creates a living record of the unit's intellectual journey. Distinguishing 'evidence' from 'new questions' teaches learners that science advances through both answers and productive new questions — a key epistemic practice.	learner posts both a green and an orange sticky note. Teacher checks that green 'evidence' notes reference at least one branch of Chemistry by name and connect it to a real-world application. Red flag: learners who write vague notes like 'Chemistry is useful' — pull these learners aside at the start of the next lesson for a brief verbal probe: 'Tell me specifically HOW Chemistry is useful — which branch, doing what?'
Model Building Phase 📹 **VIDEO:** Biological redox reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook)	Learners return to their individual concept maps started in Lesson 1. Using the class concept map as a reference, each learner individually revises and expands their own map to include: (a) all 7 branches of Chemistry as nodes connected to the central 'Chemistry' node; (b) at least one Kenyan real-world example per branch written on the connecting arrow; (c) a 'Chapati Phenomenon' node connected to Biochemistry and labelled 'Maillard reaction — irreversible'; (d) at least	Say: 'Open your concept maps from Lesson 1. You have 8 minutes to expand your personal model using everything we discovered today. Your map must include all 7 branches, at least one Kenyan example each, the Maillard reaction node, and two connections between branches that YOU choose. Add a legend box at the bottom: What can my model explain now?' [Set an 8-minute timer.] Circulate and crouch beside 4–5 learners to ask: 'Walk	Iterative Personal Model Revision with 'What Can My Model Explain Now?' Legend. Model revision is a core NGSS Science and Engineering Practice (SEP 2: Developing and Using Models). Requiring learners to articulate what their model CAN explain (rather than just what it shows) develops metacognitive awareness of the difference between description and explanation — a critical distinction in scientific thinking. The legend	Observable indicator: Expanded concept maps contain all 7 branches, at least 7 Kenyan examples, the Maillard reaction node, and 2 inter-branch connections. The 'What my model explains now' legend contains at least 2 specific statements (e.g., 'My model explains why chapati turns brown — Biochemistry, Maillard reaction' and 'My model explains why KEBS tests food — Analytical Chemistry'). Collect 5–6 maps at random to review before

Source: CK-12 🔍 Search ARES for similar readings	two inter-branch connections of their own choosing, with a label explaining the connection (e.g., 'Analytical Chemistry checks the purity of products made by Industrial Chemistry'). Learners also add a small legend: 'What my model can explain NOW' — listing 2–3 things their model can currently explain about the driving question.	me through one connection you drew between two branches — why did you draw that arrow?' [WAIT TIME: 10 seconds per learner before prompting.] With 2 minutes remaining, say: 'Finish your legend box — write at least 2 things your model can now explain.' At time, ask 2 volunteers to hold up their maps and describe one inter-branch connection they are proud of. Close by saying: 'Your model has grown from one node to a full network in just two lessons. As we gather more evidence across this unit — especially about drugs and the body — your model will grow further. The chapati on the jiko is still our anchor. Every branch we added today helps us explain one part of why that dough can never go back to being dough.'	also provides the teacher with a rapid written formative assessment artefact.	Lesson 3. Flag maps missing the Maillard connection or the legend for targeted follow-up.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. PHENOMENON CONNECTION: In the Explain phase, were learners able to connect the Maillard reaction to Biochemistry independently, or did they need significant scaffolding? What does this tell you about the depth of understanding from Lesson 1?

2. MISCONCEPTIONS: Which misconception appeared most frequently — confusing Organic Chemistry with Biology, or confusing Biochemistry with medicine? How will you address the most persistent misconception at the start of Lesson 3?

3. CONCEPT MAP QUALITY: When you reviewed the 5–6 collected maps, how many included a meaningful inter-branch connection (not just copied from the class map)? What proportion of learners reached the 'explanation' level in their legends vs. the 'description' level?

4. DQB PATTERNS: What new questions on the orange sticky notes surprised you? Were any questions about drug interactions, food safety, or nuclear medicine — topics that appear later in the unit? Plan how to validate those questions in upcoming lessons.

5. EQUITY CHECK: Which learners did NOT contribute verbally during cold-calls across all 5 phases? Identify at least 2 learners by name and design a specific low-stakes oral participation opportunity for them in Lesson 3.

6. KENYAN CONTEXT: Did learners generate their own Kenyan examples for the careers bridge activity that were not on the Branch Cards? If so, record these — they are evidence of genuine contextual transfer and can be used to enrich subsequent lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed 7 colour-coded Branch Cards, each describing one branch of Chemistry (Organic, Inorganic, Physical,

	Analytical, Biochemistry, Nuclear, Industrial) through a specific Kenyan context — including how Bamburi Cement production relates to Inorganic Chemistry, how KEBS food testing relates to Analytical Chemistry, and how the browning of chapati and nyama choma relates to Biochemistry.
What did I learn?	Learners learned that Chemistry is divided into specialised branches, each studying a different aspect of matter and its changes. Biochemistry specifically explains the Maillard reaction — the irreversible chemical transformation of proteins and sugars in chapati dough when exposed to heat — which is why a cooked chapati can never be returned to raw dough. Each branch underpins real Kenyan industries, careers, and daily-life events.
How does this explain the phenomenon?	Using their expanded concept maps and the Maillard reaction diagram, learners explained that the chapati phenomenon involves Biochemistry because it is a chemical reaction between amino acids (from wheat protein) and sugars (from wheat starch) driven by heat, producing new molecules responsible for the brown colour, aroma, and firm texture. Because new, chemically distinct molecules are formed, the change is irreversible — directly answering part of the driving question: 'Chemistry explains what happens to matter around us every day' through the lens of Biochemistry.

LESSON 3 (80 minutes): Chemistry in Agriculture and Food — How Chemistry Feeds Kenya	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explore how Chemistry underpins Kenya's agricultural and food industries, connecting chemical principles to everyday Kenyan food systems and advancing learners' ability to answer the driving question.
Knowledge	Learners explain how Chemistry is applied in Kenyan agriculture (fertilisers, pesticides, soil pH) and the food industry (preservatives, fermentation), and connect these applications to the concept of matter and its changes introduced through the chapati phenomenon.
Skills	Learners brainstorm, conduct guided research using ARES platform resources, and collaboratively construct an expanded concept map and Driving Question Board entry linking Chemistry to agriculture and food.
Attitudes	Learners demonstrate respect by acknowledging peers' contributions during group brainstorming, and show citizenship by discussing consumer rights related to food labels, preservatives, and product quality marks.
Key Inquiry Question	What is Chemistry and why is it important in our daily lives?
Purpose in Storyline	In Lessons 1 and 2, learners established that Chemistry is the study of matter and its changes, identified its branches, and began linking the chapati phenomenon to physical and chemical change. In Lesson 3, learners zoom out to explore where Chemistry operates at a national scale — in the farms of the Rift Valley and in the food products on Kenyan shelves. This expands their answer to the driving question: Chemistry does not just explain the jiko — it explains why Kenyan maize grows tall, why uji ferments, and why Pembe flour stays fresh. This evidence will be added to the DQB and concept map.
Safety Notes	No hazardous materials are used in this lesson. If printed food product labels or containers are brought in, ensure no expired or contaminated items are handled. Remind learners to wash hands after handling any packaging materials. All digital research must be conducted on the ARES platform only — learners should not navigate to external websites.

B. LESSON OVERVIEW

Learners begin by reconnecting to the chapati phenomenon — specifically the ingredients (wheat flour, water, oil, salt) — and brainstorm where those ingredients come from and what Chemistry was involved before the dough was ever made. This anchors a structured group brainstorm-and-research activity in which learner teams investigate one of four Chemistry-in-agriculture/food domains: (1) fertilisers and Kenyan maize farming in the Rift Valley, (2) pesticides and food safety, (3) soil pH and crop selection, and (4) food preservatives and fermentation (uji, Pembe flour). Teams use ARES platform resources to gather evidence, then share findings in a gallery walk. The lesson closes with learners updating the Driving Question Board and expanding their concept maps to include "Importance of Chemistry — Agriculture and Food."

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Preparation of	Learners are shown the list of chapati ingredients written on the board: wheat flour (from	Teacher writes the four ingredients on the board in large text and points to each: 'Wheat flour —	Activating Prior Knowledge + Prediction Journaling — Learners surface what they already know	Teacher circulates during individual writing and reads 3–4 books to check whether learners

[amides using DCC](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Lactic Acid Fermentation - Advanced (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	maize/wheat farms), water, cooking oil, salt. The teacher asks: 'Before any of these ingredients reached your kitchen, Chemistry was already at work. Where? How?' Learners individually write three predictions in their exercise books about how Chemistry might be involved in producing or preserving any one of these ingredients. After writing, learners share predictions with a shoulder partner for 2 minutes.	grown on a farm in Uasin Gishu County. Cooking oil — extracted from sunflower seeds in Nakuru. Salt — mined and refined. Water — treated at a water plant.' Then says: 'In your books, write THREE ways Chemistry might already be involved before this dough was ever made. You have 3 minutes — write without stopping.' [Allow 3 minutes of silent individual writing.] After writing: 'Turn to your neighbour and share one prediction each. You have 2 minutes.' [Allow 2 minutes.] Cold-call 4 learners to share predictions aloud. For each response, teacher probes: 'What branch of Chemistry from Lesson 2 might explain that?' [Allow 10-second WAIT TIME after each probe before accepting responses.] Record 4–5 distinct predictions on the board in a 'Our Predictions' column — these will be revisited at the end of the lesson.	about farming, food production, and Chemistry before receiving new information. Writing predictions first ensures all learners commit to an idea independently, preventing groupthink. Revisiting predictions at the end of the lesson creates cognitive accountability and models the scientific practice of evidence-based revision.	are connecting Chemistry (not just biology or physics) to the ingredients. Observable indicator: learner writes at least one prediction that references a chemical process (e.g., 'fertilisers add chemicals to soil', 'preservatives are chemicals that stop food going bad'). Cold-call responses reveal whether learners can name a relevant branch of Chemistry from Lesson 2.
Observe Phase 📹 **VIDEO:** [Preparation of amides using DCC](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Lactic Acid Fermentation - Advanced (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners are organised into four expert groups (approximately 8–10 learners each, depending on class size). Each group is assigned one research domain and accesses the corresponding ARES platform resource pack: Group 1 — Fertilisers and Kenyan maize farming (Rift Valley/Uasin Gishu); Group 2 — Pesticides and food safety (pyrethrum in Nakuru, aflatoxin in stored maize); Group 3 — Soil pH and crop selection (tea in Kericho — acidic soils; maize in Trans Nzoia — neutral soils); Group 4 — Food preservatives and fermentation (sodium benzoate in Pembe flour products, lactic acid fermentation in uji and fermented milk/mursik). Each group reads their resource,	Teacher assigns groups and displays the four domain titles on the board. Distribute or confirm access to ARES resource packs. Say: 'You are now a Chemistry expert team. Your job is NOT just to read — your job is to find EVIDENCE that answers our driving question: How does Chemistry explain changes to matter around us every day?' [Allow 18–20 minutes for group research and Evidence Capture Sheet completion.] Teacher circulates to each group at least twice: First visit (minutes 3–6): Check that groups are reading actively, not passively. Prompt Group 1: 'What chemical reaction happens when nitrogen fertiliser meets the soil?' Prompt Group 4:	Jigsaw Expert Groups + Evidence Capture Sheets — Each group becomes the classroom expert on one domain, ensuring breadth of coverage without cognitive overload. The Evidence Capture Sheet imposes a structured sensemaking routine: learners must name the chemical, describe what it does to matter (connecting to the unit's core idea), and explicitly link to the phenomenon. This prevents surface-level reading and forces analytical processing. The product label examination embeds the KICD PCI on consumer protection — learners practice reading real-world chemical information as informed citizens.	Teacher reads Evidence Capture Sheets during circulation. Observable indicators: (1) Group correctly names at least one real chemical (e.g., urea/NPK for Group 1, pyrethrin for Group 2, pH value for Group 3, sodium benzoate or lactic acid for Group 4). (2) Group's third column contains a sentence that uses the word 'change', 'matter', or 'reaction' — linking to the unit's core concept. (3) Group can identify at least one item on a food product label (preservative name, expiry date, or quality mark).

	completes a structured Evidence Capture Sheet (provided on ARES) with three columns: 'What chemical is involved?', 'What does it do to matter?', 'How does this connect to the chapati phenomenon or our daily food?' Groups also examine images of real Kenyan product labels (Pembe flour, Bidco oil, Kensalt) displayed on the ARES platform and identify: quality marks, expiry dates, preservative names, and caution messages.	'What is the name of the acid produced during uji fermentation — and is that a chemical change?' [Allow 10-second WAIT TIME after each prompt.] Second visit (minutes 12–16): Check Evidence Capture Sheets. Prompt: 'Which column is hardest to fill in — and why?' If groups struggle with the third column (connection to chapati/phenomenon), prompt: 'When chapati dough bakes on the jiko, something irreversible happens. Is anything similar happening here with your chemical?' At the 18-minute mark, say: 'Stop research. You have 4 minutes to prepare a 90-second group summary to share in our gallery walk. Decide: who speaks, what is your strongest piece of evidence, and what question does your evidence raise?'		
Explain Phase 📹 **VIDEO:** Preparation of amides using DCC Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Lactic Acid Fermentation - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups conduct a Gallery Walk. Each group has 2 minutes to present their 90-second summary to the class while the teacher displays their Evidence Capture Sheet on the ARES classroom screen. After all four presentations, the teacher leads a whole-class Claim–Evidence–Reasoning (CER) discussion: 'What is our class claim about the importance of Chemistry in agriculture and food? What evidence from our four groups supports it? What reasoning connects the evidence to the claim?' Learners then individually write one CER statement in their books connecting their group's finding back to the chapati phenomenon.	During Gallery Walk presentations: Stand to the side and allow each group's speaker to address the class. After each presentation, cold-call ONE learner from a different group: 'In one sentence, what was the most surprising chemical fact from that group?' [Allow 10-second WAIT TIME.] After all four presentations, draw a large CER frame on the board with three columns: CLAIM / EVIDENCE / REASONING. Ask the class: 'Based on all four groups — complete this sentence: Chemistry is important in agriculture and food because...' [Allow 15-second WAIT TIME for silent thinking before taking responses.] Cold-call 3 learners and build the class CER statement together on the board. Then say: 'Now, in your books, write YOUR OWN CER statement — but your	Claim–Evidence–Reasoning (CER) Framework — CER is a disciplinary literacy strategy rooted in NGSS Science and Engineering Practice 6 (Constructing Explanations). By requiring learners to distinguish between a claim (what they assert), evidence (what the data shows), and reasoning (why the evidence supports the claim), the strategy builds scientific argumentation skills. Connecting the reasoning explicitly to the chapati phenomenon ensures that new knowledge about agriculture and food is not isolated but woven into the unit's conceptual thread.	Teacher reads individual CER statements during writing. Observable indicators: (1) Claim is a complete, assertive statement about Chemistry's importance (not a question or a vague phrase). (2) Evidence names a specific chemical or process from their group's research. (3) Reasoning contains the word 'change', 'matter', 'irreversible', or 'chemical reaction', AND references the chapati phenomenon or the jiko. Any learner whose reasoning does not connect to the phenomenon is given the prompt: 'How is this similar to what happens to the dough on the jiko?'

		reasoning must include a connection to our chapati phenomenon. How is what your expert group found similar to or different from what happens when dough becomes chapati on the jiko?' [Allow 5 minutes for individual writing.] Cold-call 2 learners to read their CER reasoning aloud. Acknowledge strong reasoning with: 'Notice how [learner's name] connected fermentation to the irreversible change in chapati — that is exactly the kind of thinking Chemistry requires.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Preparation of amides using DCC Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Lactic Acid Fermentation - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board displayed at the front of the room. Each group writes one new sticky note: a QUESTION their research raised that is not yet answered (e.g., 'If fertilisers are chemicals, can they ever be dangerous to the food we eat?', 'Is fermentation in uji a chemical or physical change?', 'Why does soil pH affect crop growth at a molecular level?'). Learners also mark any previously posted questions (from Lessons 1 and 2) that their research today has partially or fully answered, writing 'EVIDENCE — L3' beside them.	Distribute sticky notes (one per group). Say: 'Look at the DQB. Your job is two-fold. First — write ONE new question your research today has raised. A good question starts with HOW or WHY and involves chemistry of matter. Second — walk to the board and find any question from Lesson 1 or 2 that your evidence today helps answer. Write EVIDENCE — L3 beside it.' [Allow 5 minutes.] After sticky notes are posted, teacher reads 2–3 new questions aloud and says: 'These are the questions we will keep investigating. Notice how every question connects back to the same puzzle — the chapati on the jiko. Chemistry in farming, Chemistry in food preservation — they all involve matter changing.' Point to the phenomenon image displayed on the board: 'The dough changed irreversibly. The fertiliser changes the soil. The lactic acid changes the uji. Chemistry is the science of ALL of these changes.'	Driving Question Board (DQB) Annotation — The DQB is a running public record of the class's growing understanding, modelled on the OpenSciEd instructional approach. Annotating existing questions with 'EVIDENCE — L3' makes learning progression visible to all learners and reinforces the idea that science builds incrementally. Posting new questions models scientific curiosity and the NGSS practice of Asking Questions (SEP 1). The ritual of returning to the DQB every lesson builds metacognitive habit — learners track not just what they know but what they still need to figure out.	Teacher reads new sticky note questions for quality. Observable indicator: question is genuinely chemistry-focused (mentions a chemical, a change to matter, or a mechanism) rather than a general life question. Teacher notes which groups posted questions that bridge to upcoming lessons (e.g., drug interactions, molecular structure) — these will be referenced in Lessons 4–5.
Model Building	Learners open their personal concept maps (started in Lesson 1	Say: 'Open your concept maps. In Lesson 1 we placed Chemistry at	Cumulative Concept Mapping — Concept mapping operationalises	Teacher scans concept maps during circulation. Observable

Phase 📹 **VIDEO:** Preparation of amides using DCC Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Lactic Acid Fermentation - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and expanded in Lesson 2). They add a new node: 'Importance of Chemistry — Agriculture and Food' connected to the central node 'Chemistry: the study of matter and its changes.' From this new node, they draw at least FOUR sub-nodes representing specific examples from today's lesson (e.g., NPK fertilisers → soil chemistry → higher maize yield; lactic acid fermentation → uji; sodium benzoate → Pembe flour preservation; soil pH → tea farming in Kericho). Each connection must be labelled with a linking phrase that describes the relationship (e.g., 'uses', 'causes a change in', 'prevents', 'determines'). Learners also add a small sketch or symbol connecting today's additions back to the chapati phenomenon node already on their map.	the centre. In Lesson 2 we added the branches. Today we add a whole new branch: Chemistry in Agriculture and Food.' Model on the board by drawing the new node and ONE example sub-node with a labelled arrow. Then say: 'You need at least FOUR sub-nodes from today's lesson, each with a labelled arrow. AND you must draw a connection — even a dotted line — from this new branch back to the chapati phenomenon node. Why? Because ALL of these are examples of Chemistry explaining changes to matter in our daily lives.' [Allow 8–10 minutes.] Circulate and look at concept maps. Stop beside 2–3 learners and cold-call them to explain one connection aloud: 'Tell me in one sentence what this arrow means — what is the relationship between soil pH and tea farming in Kericho?' [Allow 10-second WAIT TIME.] Close the lesson: 'By Lesson 10, your concept map will be a complete picture of how Chemistry explains the world around us — from the chapati on the jiko, to the farm, to the pharmacy. Keep it safe.'	NGSS Science and Engineering Practice 2 (Developing and Using Models). By building the same map across all 10 lessons, learners create a visual model of growing disciplinary understanding rather than a collection of isolated facts. The requirement to label every arrow with a relational phrase (not just draw lines) forces propositional thinking — learners must articulate the nature of each relationship, not just its existence. Connecting the new branch to the chapati phenomenon node ensures coherence across the unit's storyline.	indicators: (1) New 'Agriculture and Food' node is present and connected to the central Chemistry node with a labelled arrow. (2) At least three of the four required sub-nodes are present with labelled arrows. (3) A connection (even dotted/annotated) links today's content back to the chapati/phenomenon node. Learners whose maps lack the phenomenon connection are asked: 'How is NPK fertiliser changing soil similar to what heat did to the chapati dough on the jiko?'

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did learners' CER reasoning statements successfully connect the agriculture/food Chemistry evidence back to the chapati phenomenon — or did most learners treat the phenomenon and today's content as separate topics? If the connection was weak, plan a 5-minute 'phenomenon bridge' opener for Lesson 4. (2) Which of the four expert groups produced the most analytically rich Evidence Capture Sheet? What made their research process different — group composition, reading strategy, or the richness of the ARES resource? Use this to inform grouping for Lesson 4. (3) Review the new DQB sticky note questions. Are any questions pointing toward drug and substance use (e.g., 'If fertilisers are chemicals, can chemicals in drugs be dangerous too?')? Flag these for the Lesson 6–7 pivot to the drug and substance use thread of the unit. (4) Did the product label examination (Pembe flour, Bidco oil, Kensalt) generate genuine consumer awareness discussion — or was it rushed? If rushed, embed a 3-minute label-reading routine into Lesson 4's opener. (5) Core competency check: Did learners demonstrate Communication and Collaboration through genuine group decision-making, or did one or two voices dominate each group? Consider assigning rotating roles (Reader, Recorder, Speaker, Questioner) in Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined ARES platform resources on fertilisers (NPK/urea) used in Rift Valley maize farming, pesticides (pyrethrum) in Nakuru, soil pH effects on tea in Kericho and maize in Trans Nzoia, and food preservatives (sodium benzoate) and fermentation (lactic acid in uji and mursik). They also observed real Kenyan food product labels (Pembe flour, Bidco oil, Kensalt) identifying preservatives, expiry dates, quality marks, and caution messages.
What did I learn?	Chemistry is applied in Kenyan agriculture through fertilisers that change soil chemistry to support crop growth, pesticides that target harmful organisms through chemical reactions, and soil pH management that determines which crops thrive in which regions. In the food industry, Chemistry explains how preservatives prevent spoilage by inhibiting microbial chemical reactions and how fermentation — a chemical change — transforms maize porridge into uji and milk into mursik. These are all examples of Chemistry explaining and controlling changes to matter in daily Kenyan life.
How does this explain the phenomenon?	Just as the chapati dough undergoes an irreversible chemical change on the jiko — transforming matter through heat — Chemistry operates in every stage of Kenya's food chain: changing soil to grow the wheat, preserving the flour on the shelf, and fermenting the uji in the pot. Understanding these chemical changes helps Kenyans make informed, safe, and smart choices — as farmers, consumers, and citizens.

LESSON 4 (80 minutes): Chemistry in Medicine, Pharmaceuticals, and Manufacturing	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how Chemistry is applied in medicine, the pharmaceutical industry, and manufacturing — and to connect those applications back to the anchoring phenomenon of chapati dough transforming on the jiko.
Knowledge	Learners will be able to: (a) explain how Chemistry underpins medical diagnostics, anaesthetics, and treatment; (b) describe how Chemistry drives drug synthesis, quality control (KEBS standards), and pharmaceutical manufacturing in Kenya; (c) identify chemical processes used in Kenyan manufacturing industries such as EABL and Bidco; (d) explain the meaning of the terms drug, prescription, and dosage, and examine the effects of drug and substance use in day-to-day life.
Skills	Learners will be able to: (a) extract and synthesise key evidence from an informational reading text; (b) update a concept map with new evidence from multiple Chemistry application sectors; (c) compose and defend a concise evidence statement linking Chemistry to real-world impact; (d) evaluate the role of KEBS and pharmaceutical quality control as safeguards for consumers.
Attitudes	Learners will: (a) appreciate the life-saving role of Chemistry in medicine and pharmaceuticals; (b) demonstrate respect for peers' ideas during group discussions (Value: Respect); (c) advocate for responsible drug use and a safe, healthy learning environment (PCI: Health Promotion — drug and substance use); (d) show awareness of consumer rights when examining product labels (PCI: Socio-Economic and Environmental issues — consumer protection).
Key Inquiry Question	What is Chemistry and why is it important in our daily lives? / How does drug and substance use affect our health and well-being?
Purpose in Storyline	In Lessons 1–3, learners established what Chemistry is, explored its branches, and examined its role in agriculture and food. Lesson 4 advances the evidence-gathering sequence by moving into the human body and industry — revealing that the same chemical changes seen in chapati on the jiko (irreversible reactions, energy transfer, new substance formation) also occur inside us when we take medicines, and inside factories when products like cooking oil and beverages are manufactured. This lesson is the crucial bridge between 'Chemistry in the kitchen' and 'Chemistry as a life-or-death discipline,' directly setting up the drug and substance use discussions mandated by KICD.
Safety Notes	No hazardous materials are used in this lesson. If physical medicine containers or product labels are brought in, remind learners not to open, taste, or inhale any substances. Emphasise that medicines must only be used as prescribed by a qualified health professional. Sensitivity note: some learners may have personal or family experience with drug and substance abuse — frame all discussions with empathy and without stigma.

B. LESSON OVERVIEW
Learners begin by revisiting the chapati phenomenon and their Driving Question Board to identify gaps in their understanding about how Chemistry affects the human body and industries. They make predictions about what happens chemically when a painkiller (e.g., paracetamol) enters the body. The core evidence activity is a structured reading on Chemistry in medicine, pharmaceuticals, and manufacturing, set in Kenyan contexts (Kenyatta National Hospital blood tests, KEBS pharmaceutical standards, EABL fermentation, Bidco oil refining). Using a Jigsaw reading strategy, groups become 'sector experts' and share findings with peers. Learners then apply evidence to explain the chapati phenomenon connection, update their concept maps, and post new questions and answers on the Driving Question Board. The lesson closes with a brief formative exit task on drug terminology (drug, prescription, dosage) aligned to KICD PCIs.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Debt loops rationale and effects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a split image: (Left) a ball of raw chapati dough on a jiko; (Right) a packet of paracetamol tablets beside a diagram of the human digestive system. The teacher asks: 'Both images involve Chemistry. In your exercise book, write TWO predictions: (1) What chemical change do you think happens inside your body when you swallow a paracetamol tablet? (2) What do you think would happen if you took five times the recommended dose?' Learners write individual predictions for 3 minutes, then share with a shoulder partner. Three pairs are cold-called to share. Predictions are NOT corrected — they are posted on a 'Before Evidence' column on the Driving Question Board for revisiting at the end of the lesson.	Display the split image prominently (printed A3 poster or projected). Say: 'Look at both images carefully. Think about what we already know — Chemistry is the study of matter and its changes. Both the chapati and this tablet involve chemical changes.' Pause. Say: 'In your exercise book, write your two predictions now. You have 3 minutes — no talking yet.' [START TIMER — 3 minutes silent individual writing.] After 3 minutes: 'Turn to your shoulder partner. Share your predictions. You have 90 seconds.' [WAIT — 90 seconds.] Cold-call 3 pairs using name sticks. For each pair say: 'Tell us your prediction for Question 1.' Do NOT evaluate — say: 'Thank you — we will test that prediction against evidence today.' Write 2–3 student predictions verbatim on the 'Before Evidence' column of the DQB. Say: 'Keep those predictions in mind — our reading today will help us decide if they are supported by evidence.'	Strategy: Predict-Before-Evidence (Anchored Prediction). By connecting the familiar chapati phenomenon to a new context (medicine), learners activate prior knowledge and surface misconceptions (e.g., 'the tablet just dissolves and disappears'). Writing predictions before reading creates cognitive dissonance that motivates careful reading — a core NGSS Science and Engineering Practice (Obtaining, Evaluating and Communicating Information).	Observable indicator: Every learner has two written predictions in their exercise book. Teacher circulates during the 3-minute writing window and scans books — look for learners who are blank (prompt: 'Think about what we said in Lesson 1 — what does Chemistry study?'). Listen during cold-calls for common misconceptions about medicine (e.g., 'tablets are not chemicals') — note these for targeted questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** Debt loops rationale and effects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12	Learners engage in a Jigsaw Reading activity using a structured 2-page reading titled 'Chemistry at Work: Medicine, Pharmaceuticals and Manufacturing in Kenya.' The reading has three clearly labelled sections: Section A — Chemistry in Medicine (blood tests at KNH, anaesthetics in surgery, how paracetamol works at the molecular level); Section B — Chemistry in the Pharmaceutical Industry (drug synthesis, meaning of drug/prescription/dosage, KEBS standards, role of quality control);	Distribute the reading (or display on ARES platform). Say: 'This is our evidence source for today — a reading about how Chemistry is used in real Kenyan contexts. You will become an expert in one section.' Assign sections clearly by seat position. Say: 'Read ONLY your assigned section. As you read, annotate three things: the chemical change described, how Chemistry protects people, and one new word. You have 8 minutes — begin now.' [START TIMER — 8 minutes. Circulate —	Strategy: Jigsaw Reading with Expert Teaching. This strategy mirrors the NGSS practice of Obtaining, Evaluating and Communicating Information while also developing the KICD core competency of Communication and Collaboration. By teaching peers, learners must process information deeply rather than passively. The Evidence Table explicitly requires learners to connect every new piece of information back to the chapati phenomenon — preventing	Observable indicator: Every learner has a partially or fully completed 3-column Evidence Table. During Expert Group circulation, listen for: (1) correct use of 'drug,' 'prescription,' and 'dosage' in Section B groups — if absent, note for targeted questioning in Phase 3; (2) whether Section C groups can name fermentation as a chemical change — this checks retention from Lesson 3 (agriculture/food). Cold-call answers reveal whether the chapati connection is being

🔍 Search ARES for similar readings	Section C — Chemistry in Manufacturing (EABL fermentation chemistry, Bidco palm oil refining, quality marks on product labels). In Home Groups (4 learners each), each learner is assigned one section (one learner covers two short sub-sections). Learners read their assigned section independently for 8 minutes, annotating: (1) one chemical change described, (2) one way Chemistry protects people, (3) one new vocabulary word. Learners then move into Expert Groups (all Section A readers together, all Section B readers together, all Section C readers together) to discuss their section for 5 minutes and agree on the 3 most important points to teach their Home Group. Learners return to Home Groups and each expert teaches their section (2 minutes per expert). All learners complete a 3-column Evidence Table in their exercise books: Sector | Chemical Change or Process | How It Connects to the Chapati Phenomenon.	do not explain content yet. If a learner asks a comprehension question, respond: 'Mark it with a question mark — bring it to your Expert Group.'] After 8 minutes: 'Expert Groups — move now. All Section A readers, go to the front-left cluster. Section B, front-right. Section C, back cluster.' [Allow 60 seconds for movement.] Say: 'In your Expert Group, discuss your section. Agree on the 3 most important points. You have 5 minutes.' [TIMER — 5 minutes. Circulate to Expert Groups. Prompt Section B groups: 'Can you define the word dosage in your own words?' Prompt Section C groups: 'What chemical process does EABL use — and have you seen that word before in our lessons?'] After 5 minutes: 'Return to your Home Groups now.' [60 seconds movement.] Say: 'Each expert has exactly 2 minutes to teach your section. The listeners fill in the Evidence Table as they hear the information. Expert A — begin.' [Use a visible 2-minute timer for each expert turn.] After all teaching: 'You now have 3 minutes to complete any gaps in your Evidence Table individually.' Cold-call 3 learners (one per section) to share one row of their Evidence Table with the class. Write key terms on the board: DRUG | PRESCRIPTION | DOSAGE | KEBS | FERMENTATION | QUALITY CONTROL.	isolated fact-learning and keeping the anchoring phenomenon central. Requiring learners to identify 'how Chemistry protects people' in each sector plants the seed for the drug and substance use discussion in Phase 3.	made: a strong answer sounds like 'In both chapati-making and drug synthesis, new substances are formed that cannot be reversed.'
Explain Phase 📹 **VIDEO:** Debt loops rationale and effects Source: Khan Academy (English	Learners use their Evidence Table to answer three targeted sensemaking questions, first individually (3 minutes), then in their Home Groups (4 minutes), then in a whole-class share-out. The three questions are: (Q1) 'A	Write the three questions on the board or display on ARES. Say: 'Using your Evidence Table and what you heard from your experts, answer all three questions individually first. You have 3 minutes — write in full sentences.'	Strategy: Structured Academic Controversy into Direct Instruction Bridge. The three-tiered questions move learners from recall (Q1 — vocabulary application) to analysis (Q2 — quality control reasoning) to synthesis (Q3 — phenomenon	Observable indicator: Q1 answers must correctly embed all three vocabulary words (drug, prescription, dosage) in a meaningful sentence — teacher marks these during circulation. Q3 answers must name a specific

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	doctor at Kenyatta National Hospital prescribes paracetamol to a patient. Using the words drug, prescription, and dosage correctly in one sentence, explain why the patient must follow the doctor's instructions exactly.' (Q2) 'KEBS inspects a batch of tablets manufactured at a Nairobi pharmaceutical factory and rejects it. What chemical reason might explain the rejection — and why does this matter for Kenyan consumers?' (Q3) 'The chapati dough on the jiko undergoes an irreversible chemical change. Identify ONE chemical change from today's reading (medicine, pharmaceuticals, or manufacturing) that is also irreversible, and explain why it cannot be reversed.' After the group discussion, a whole-class Gallery-Walk-style share-out occurs: one spokesperson per Home Group stands and delivers a 30-second answer to one assigned question. The teacher facilitates, probes, and corrects misconceptions. The teacher then delivers a 5-minute direct-instruction close on: (a) how molecular-level changes in drug synthesis parallel the dough-to-chapati transformation; (b) the dangers of incorrect dosage and self-medication; (c) the role of KEBS as Kenya's chemical quality guardian.	[TIMER — 3 minutes. Circulate — scan for Q1 answers that misuse vocabulary; for Q3, look for learners who have not identified an irreversible change.] After 3 minutes: 'Discuss with your Home Group. Do you agree? Can you improve your answers? 4 minutes.' [TIMER — 4 minutes.] For the share-out: assign Q1 to Groups 1 & 2, Q2 to Groups 3 & 4, Q3 to Groups 5 & 6 (adjust for class size). Say: 'Spokesperson for Group 1 — stand and give us your answer to Q1 in 30 seconds.' After each answer, ask the class: 'Does anyone want to add or challenge that answer?' [WAIT TIME — 10 seconds before cold-calling.] Cold-call at least 2 additional learners per question. For Q3, specifically probe: 'Why is it irreversible? What happened to the original molecules?' If a learner says 'the tablet dissolves,' gently redirect: 'Dissolving is a physical change — what did the reading say happens to the active ingredient at the molecular level?' For direct instruction close, stand at the board and say: 'Let me now connect everything together.' Draw a two-column table: LEFT = Chapati on Jiko | RIGHT = Paracetamol in Body. Fill in: raw material → product, irreversible change, new substance formed, energy involved. Say: 'This is why Chemistry matters — it explains changes in the kitchen AND in the human body. But here is the critical point —' [pause 5 seconds] '— what happens if you put too much heat on the chapati?' [Cold-call 2 learners.] 'Exactly. It burns — it is destroyed. The same is true in the body with an incorrect	connection). The direct-instruction bridge at the end of Phase 3 ensures that the sensemaking is consolidated and misconceptions are resolved before learners update their models. The chapati-paracetamol analogy (both involve irreversible molecular changes) is the conceptual anchor that ties the lesson's evidence back to the phenomenon — a hallmark of storyline-based teaching.	irreversible chemical change from the reading (not just 'mixing' or 'dissolving') — the teacher scans for this during group discussion. During the share-out, listen for the phrase 'new substance formed' or 'cannot be reversed' as indicators of deep understanding. If fewer than 50% of groups achieve this in Q3, pause and revisit the chapati phenomenon explicitly: 'Remember Day 1 — we asked can we turn the chapati back to dough? Why not?'

		dosage. Chemistry tells us why.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Debt loops rationale and effects Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. The teacher reads aloud the three evidence statements posted on the DQB at the start of the unit (from the Teacher Template): (1) 'Chemistry helps us understand how drugs interact with the body at a molecular level and helps in recognising the risks of drug abuse and developing safe medications.' (2) 'Matter is anything that has mass and takes up space — studying Chemistry helps us understand the properties, composition, and changes of matter.' (3) Evidence from Lesson 3 on states of matter. Learners are given two sticky notes each. On the YELLOW sticky note, they write one new answer or piece of evidence from today's lesson that helps answer the driving question: 'How does Chemistry explain what happens to the matter around us every day?' On the BLUE sticky note, they write one new question that today's lesson has raised for them. Learners post their sticky notes on the DQB. The teacher reads out 4–5 sticky notes aloud and facilitates a 3-minute discussion: 'Which of today's new questions do you think we will need Chemistry's deeper tools — like understanding atoms and molecules — to answer?'	Stand at the DQB. Say: 'Let's see how our understanding has grown since Lesson 1.' Read the three evidence statements from the template aloud slowly. Say: 'Today you gathered new evidence. I want you to think: what has changed in your understanding?' Distribute yellow and blue sticky notes. Say: 'On the YELLOW note — write one new piece of evidence from today that answers any part of our driving question. Be specific — use a word from today's lesson. On the BLUE note — write a new question you now have. You have 2 minutes.' [TIMER — 2 minutes, silent.] Say: 'Come and post your notes now.' [Allow 90 seconds for posting.] Read 4–5 yellow notes aloud: 'Interesting — this learner says...' Then read 2–3 blue notes. Ask the class: 'This blue question asks [read question]. To answer this, what deeper knowledge do you think we will need?' [WAIT TIME — 12 seconds.] Cold-call 3 learners. Guide responses toward: 'We will need to understand atoms and molecules — and that is exactly where we are headed in Sub-Strand 1.2.' This plants the transition to the next sub-strand organically.	Strategy: Driving Question Board (DQB) — Evidence Accumulation and Question Generation. The DQB is a living public record of the class's growing understanding, directly modelled on the NGSS storyline approach. By colour-coding answers (yellow) and new questions (blue), learners visually see that good science generates as many new questions as it answers. The teacher's closing prompt — connecting blue questions to the atomic structure sub-strand — demonstrates how each lesson is a stepping stone in a coherent explanatory sequence, reinforcing the KICD storyline thread.	Observable indicator: Every learner posts both a yellow and a blue sticky note. Quality check — a strong yellow note references a specific term from today (e.g., 'KEBS ensures drugs are chemically safe for Kenyans') rather than a vague statement ('Chemistry is important'). A strong blue note asks a mechanistic question (e.g., 'How does paracetamol's molecule know which pain signal to block?') rather than a recall question. Teacher photographs the DQB at the end of the lesson for tracking progression across all 10 lessons.
Model Building Phase 📹 **VIDEO:** Debt loops rationale and effects Source: Khan	Learners open their running Concept Map (begun in Lesson 1 and updated each lesson). They add a new branch titled 'Chemistry in Medicine, Pharma & Manufacturing' and connect at least FOUR new nodes using today's evidence: (1) Drug /	Say: 'Open your concept maps. We are going to add today's evidence — four new branches minimum.' Display a partially completed model on the board (or ARES screen) with the central node 'Chemistry' and the chapati phenomenon node already present	Strategy: Cumulative Concept Map Revision with Phenomenon Connection Statement. The concept map is the learner's personal running model of Chemistry — it grows with each lesson, making learning visible and cumulative. Requiring a labelled	Observable indicator: Concept maps must show at least 4 new labelled nodes with relationship arrows. Teacher collects 6–8 concept maps at random for review after the lesson (rotate across the class across the 10-lesson unit). Phenomenon

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Prescription / Dosage → links to 'Chemistry in the Body'; (2) KEBS Quality Control → links to 'Consumer Protection'; (3) Fermentation (EABL) / Oil Refining (Bidco) → links to 'Chemical Changes in Industry'; (4) Irreversible Chemical Change → links back to the central phenomenon node 'Chapati on Jiko.' Learners draw labelled arrows showing the relationship between nodes (e.g., 'KEBS ensures → safe dosage → protects consumers'). Learners then write a two-sentence 'Phenomenon Connection Statement' at the bottom of their concept map page: 'Today's evidence helps explain the chapati phenomenon because ______. This shows that Chemistry is important in daily life because ______.' Learners share their Phenomenon Connection Statement with their shoulder partner and give each other one star (something strong) and one step (something to improve).	from previous lessons. Say: 'Your first new branch is called Chemistry in Medicine, Pharma and Manufacturing — write that now.' Walk around and check that learners have started the branch. Say: 'Now add at least four nodes from today's lesson. Use your Evidence Table to help you. Label every arrow with a relationship word — like causes, ensures, prevents, produces. You have 5 minutes.' [TIMER — 5 minutes. Circulate — prompt learners who have fewer than 4 nodes: 'What did Section B experts teach you about KEBS? Add that.' Prompt advanced learners: 'Can you draw a connection between the irreversible chemical change in pharma and the irreversible change in chapati?'] After 5 minutes: 'Now write your two-sentence Phenomenon Connection Statement at the bottom of the page. You have 90 seconds.' [TIMER.] Say: 'Share with your shoulder partner. Give one star and one step.' [90 seconds peer feedback.] Cold-call 2 learners to read their Phenomenon Connection Statement to the class. Affirm strong connections. Close the lesson by saying: 'Look at your concept map. In Lesson 1, it had very few branches. Now look at how much Chemistry explains. Next lesson, we go further — but the chapati is still our anchor. We are still asking: what exactly changed in that dough at the level we cannot see?'	relationship arrow (not just connected nodes) forces learners to articulate the mechanism, not just the association — a direct implementation of the NGSS practice of Developing and Using Models. The Phenomenon Connection Statement ensures the model remains tethered to the anchoring phenomenon and does not become an abstract vocabulary list. Peer feedback (star and step) develops the KICD core competency of Communication and Collaboration.	Connection Statements are scanned during cold-call — a strong statement explicitly names a chemical change from today's reading AND links it to the chapati phenomenon. Weak statements ('Chemistry is used in medicine') are flagged for follow-up at the start of Lesson 5. Exit check: before learners leave, ask one whole-class question: 'Raise your hand if you can name one product made in Kenya whose quality is protected by Chemistry.' Expect at least 70% of hands — if fewer, note this as a starting point for Lesson 5 recap.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. PHENOMENON CONNECTION: When learners wrote their Phenomenon Connection Statements, did the majority explicitly link an industrial or medical chemical change to the chapati transformation — or did they treat medicine and manufacturing as separate topics? If the connection was weak, plan a 5-minute recap at the start of Lesson 5 using the two-column board diagram (Chapati on Jiko vs. Paracetamol in Body).

2. VOCABULARY DEPTH: Did learners use drug, prescription, and dosage correctly and confidently in Q1 answers, or were the words used loosely? If vocabulary depth was insufficient, consider a vocabulary reinforcement strategy in Lesson 5 (e.g., a quick Frayer Model for 'dosage').

3. JIGSAW QUALITY: Were Expert Group discussions substantive — did experts arrive at Home Groups ready to teach, or did they struggle? If Expert Groups were underprepared, review the reading text for accessibility and consider providing a 3-bullet summary scaffold for weaker readers in the next Jigsaw activity.

4. DQB QUALITY: Review the blue sticky notes photographed from the DQB. Count how many ask mechanistic questions (about molecules, atoms, reactions) versus surface questions (about facts). A high proportion of mechanistic questions indicates learners are ready for Sub-Strand 1.2 (The Atom). A high proportion of factual questions suggests the storyline thread needs reinforcement.

5. INCLUSIVITY: Were all learners actively engaged during Expert Group and Home Group phases — particularly girls and learners who are quieter? Note any learner who did not post a sticky note or whose concept map remained incomplete. Plan a targeted check-in at the start of Lesson 5.

6. DRUG AND SUBSTANCE USE SENSITIVITY: Did any discussion moment require sensitive facilitation (e.g., a learner disclosing personal experience with drugs)? If so, note the approach used and whether a referral to the school counsellor may be appropriate.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners read and analysed a structured text on Chemistry in medicine (KNH blood tests, anaesthetics, paracetamol), the pharmaceutical industry (drug synthesis, KEBS standards, quality control), and manufacturing (EABL fermentation, Bidco oil refining). They observed how Chemistry operates as the underlying science in each sector and examined product labels for quality marks, expiry dates, and dosage instructions.
What did I learn?	Learners learned that: (1) Chemistry explains how drugs interact with the body at a molecular level, including why incorrect dosage causes harm; (2) the terms drug, prescription, and dosage have precise scientific meanings that protect consumers; (3) KEBS uses chemical quality standards to safeguard Kenyan consumers; (4) manufacturing industries like EABL (fermentation) and Bidco (oil refining) depend on controlled chemical changes — the same type of irreversible change seen when chapati dough is placed on the jiko.
How does this explain the phenomenon?	Learners explained how the irreversible chemical change in chapati-making on the jiko is conceptually identical to chemical changes in pharmaceutical manufacturing and in the body — in all cases, new substances are formed at the molecular level and the original matter cannot be recovered. They explained that Chemistry is therefore not only a kitchen science but a life-protecting discipline that underpins medicine, consumer safety, and industry in Kenya.

LESSON 5 (80 minutes): Chemistry in Other Industries: Nuclear, Sports, Entertainment & Energy	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explore the role of Chemistry in the nuclear, sports, entertainment, and energy industries, and connect these roles back to the anchoring phenomenon of matter change.
Knowledge	Learners will be able to identify and explain how Chemistry operates in nuclear energy (fission/fusion, Kenya's energy mix), sports nutrition (supplements, hydration science for Kenyan marathon runners), entertainment (dyes, pigments, special effects), and environmental/energy chemistry.
Skills	Learners will practise research, collaborative discussion, and oral presentation skills by gathering, synthesising, and presenting evidence on Chemistry's industrial roles.
Attitudes	Learners will appreciate that Chemistry is not confined to a laboratory — it underpins decisions that affect real Kenyan lives, from what athletes consume to how electricity is generated and how entertainers create colour and spectacle.
Key Inquiry Question	What is Chemistry and why is it important in our daily lives?
Purpose in Storyline	In Lessons 1–4 learners defined Chemistry, mapped its branches, and traced its role in agriculture, food, pharmaceuticals, and medicine. Lesson 5 widens the lens further — into nuclear energy, sports science, entertainment, and environmental chemistry — so learners build a panoramic view of Chemistry at work. Every industry discussed links back to the anchoring phenomenon: matter changes (physical or chemical) are at the heart of how a nuclear reactor, a runner's energy gel, a stage pyrotechnic, and a solar cell all function. This evidence equips learners to answer the driving question with increasing depth, preparing them for the drug-and-substance-use focus in Lessons 6–8.
Safety Notes	No hazardous materials are used in this lesson. If internet-connected devices are available, remind learners to access only approved, school-safe websites. Discussions about nuclear energy should be kept factual; teacher should be prepared to address misconceptions about radiation calmly and accurately. No physical experiments are conducted today.

B. LESSON OVERVIEW
Learners work in four expert research groups (Nuclear & Energy; Sports & Nutrition; Entertainment & Dyes; Environmental Chemistry), each investigating how Chemistry drives their assigned industry using print/digital resources provided by the teacher. Groups prepare a 3-minute plenary presentation using a poster or mini-whiteboard. After presentations, the class uses a Jigsaw debrief to cross-pollinate knowledge. Learners then update the Driving Question Board (DQB) and revise their cumulative class model of 'What Chemistry Does'. The lesson reinforces that every industrial application — from Kenya Power's grid to Eliud Kipchoge's race-day nutrition — involves matter changing in ways that Chemistry can explain, predict, and control, mirroring the irreversible change that turned chapati dough into a golden disc on the jiko.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	The teacher displays four images	Display the four images and say:	Think-Pair-Share with a Prediction	Observable indicator: All learners

📹 **VIDEO:** [Creativity break: How do you apply creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	on the board (or printed A3 sheets): (1) a diagram of a nuclear power station, (2) Kenyan marathon runners at the finish line in Boston with energy gels in hand, (3) a vibrant Nairobi carnival float with coloured smoke and dyed costumes, (4) a solar panel installation in Garissa. Learners work individually for 3 minutes to write in their notebooks: 'Where do you think Chemistry is hiding in each of these images? Write one specific chemical idea you think is involved.' Learners then share predictions with a partner (Think-Pair) for 2 minutes before selected pairs share with the class. Predictions are recorded on a 'Prediction Parking Lot' section of the DQB.	'Before we research anything today, I want your raw predictions. Look at these four images carefully. Where is Chemistry hiding? You have 3 minutes — write silently in your notebook.' Set a visible timer for 3 minutes and enforce silence. After 3 minutes say: 'Now turn to your partner and share your ideas. You have 2 minutes.' After pair discussion, cold-call exactly 4 pairs (one per image) using a name stick: 'Amina, what did you and your partner predict about the nuclear station?' After each response, offer a neutral prompt — 'Interesting — what makes you think that?' — and wait 10–12 seconds for elaboration. Do NOT confirm or correct predictions at this stage. Record key prediction phrases verbatim on the Prediction Parking Lot on the DQB using a different-coloured marker.	Parking Lot. By externalising predictions before research, learners activate prior knowledge and create cognitive hooks. The Parking Lot makes predictions visible so they can be revisited and revised, modelling how scientists update hypotheses with evidence — directly mirroring the unit's anchoring practice of predicting whether chapati dough can be reversed.	have written at least one prediction per image in their notebooks within the 3-minute window (teacher circulates and scans notebooks). Quality indicator: At least 3 of the 4 cold-called pairs name a specific substance or process (e.g., 'uranium', 'carbohydrates', 'pigment', 'photovoltaic cells') rather than vague answers like 'chemicals are used'. If responses are vague, teacher probes: 'Can you name a specific material or change you are thinking of?'
Observe Phase 📹 **VIDEO:** [Creativity break: How do you apply creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are organised into four expert groups of approximately equal size. Each group receives a printed Research Briefing Card (prepared by the teacher in advance) containing 3–4 paragraphs of curated, Kenya-relevant information, 2–3 guiding research questions, and a graphic organiser to fill in. Where devices are available, groups may supplement with approved websites. Groups have 18 minutes to read, discuss, and complete their graphic organiser. Group roles are assigned: Reader, Recorder, Timekeeper, Presenter. GROUP 1 — Nuclear & Energy: Reads about nuclear fission (uranium atoms splitting, chain reactions), Kenya's current energy mix (geothermal at Olkaria, hydro	Distribute Research Briefing Cards and say: 'Each group is now an expert team. Your job is to become the class experts on your industry. Read the briefing card together, fill in your graphic organiser, and prepare a 3-minute presentation. You have 18 minutes — your Timekeeper must alert the group at 10 minutes and again at 15 minutes.' Circulate to each group at least twice during research time. At Group 1, prompt: 'How is a nuclear reaction similar to — or different from — what happened to the chapati dough on the jiko?' Wait 12 seconds. At Group 2, prompt: 'Kipchoge's muscles need glucose to contract. Where does that glucose come from and what branch of Chemistry studies it?' At Group 3, prompt: 'When a copper	Expert Jigsaw with Graphic Organisers. This strategy distributes cognitive load across the class — no single learner must master all four industries. The Jigsaw Listener Sheet forces active listening and cross-industry synthesis. Requiring a connection to the chapati phenomenon in every listener note ensures the anchoring phenomenon remains the conceptual spine, preventing fragmentation of knowledge into disconnected facts.	Observable indicators: (1) Every group's graphic organiser is at least 75% complete by the end of the 18-minute research window (teacher checks during circulation). (2) Each presentation includes at least one named chemical substance or process (not just 'chemistry is used'). (3) Every learner's Jigsaw Listener Sheet has an entry for each of the four presentations. Quality indicator: Cold-called connections to the chapati phenomenon are plausible (e.g., 'Both nuclear reactions and cooking the chapati involve energy causing a change in matter that cannot be easily reversed'). If a group's presentation is too vague, teacher asks a targeted clarifying question: 'Can you give us a specific example from Kenya of

	at Masinga, KPLC grid), and the concept of nuclear power as a potential future source. Identifies chemical changes involved (nuclear reactions as extreme matter change — compare to chapati browning). GROUP 2 — Sports & Nutrition: Reads about carbohydrate loading (glycogen as stored glucose — a chemical substance in muscles), electrolyte replacement (sodium, potassium lost in sweat), and protein supplementation for muscle repair. Contextualised around Kenyan marathon runners (Eliud Kipchoge, Faith Kipyegon) and what they consume before and during a race. GROUP 3 — Entertainment & Dyes: Reads about how synthetic dyes are made from chemical reactions (chromophore groups absorbing specific light wavelengths), how special-effects pyrotechnics use metal salt combustion to produce coloured flames (copper = green, sodium = yellow — as seen in Nairobi festivals and film sets), and how food colouring in mandazi and cakes is regulated by KEBS for safety. GROUP 4 — Environmental Chemistry: Reads about how Chemistry monitors water quality in Lake Victoria, studies air pollution in Nairobi's industrial areas (Mombasa Road corridor), analyses soil chemistry in Rift Valley farms, and develops biodegradable plastics. After 18 minutes, each group prepares a 3-minute verbal presentation with a mini-poster or whiteboard summary. Groups present in sequence (12 minutes total, 3 minutes each). During each presentation, listeners complete a 'Jigsaw Listener Sheet' noting: one	compound burns green in a firework, is that a physical or chemical change? How do you know?' At Group 4, prompt: 'If a factory on Mombasa Road releases sulphur dioxide, what does Chemistry tell us about what happens to that gas in the air?' During presentations, stand to the side and do not interrupt. After each presentation, ask the listening class: 'What is one connection you wrote between this group's industry and our chapati phenomenon?' Cold-call 2 learners per presentation (8 cold-calls total). Use wait time of 10 seconds before accepting responses.		this chemical process?'

	new fact learned and one connection to the chapati phenomenon.			
Explain Phase 📹 **VIDEO:** Creativity break: How do you apply creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class synthesis discussion (10 minutes). On the board, the teacher draws a large circle labelled 'Chemistry Explains Matter Change' and four spokes labelled: Nuclear/Energy, Sports/Nutrition, Entertainment, Environment. As a class, learners contribute one key chemical idea from each group's presentation to populate each spoke. The teacher then poses the 'Big Connection Question': 'In every single industry we have discussed — nuclear, sports, entertainment, environment — what is the ONE thing Chemistry is always doing?' Learners write a solo response in their notebooks for 2 minutes. Selected learners share. The teacher guides the class to the generalisation: Chemistry is always tracking, explaining, or controlling how matter changes — just as it explains why the chapati dough changed and cannot be recovered. Learners then complete a 'Claim-Evidence-Reasoning' (CER) frame individually: Claim: 'Chemistry is essential in the [chosen industry].' Evidence: One specific fact from the research. Reasoning: How that fact connects to matter changing.	Draw the spoke diagram and say: 'Help me fill this in — one key chemical idea from each group. I want the idea in Chemistry language, not just the industry name.' Cold-call one learner per spoke (4 calls). After the diagram is populated, pause and say: 'Now here is the big question. Look at all four spokes. What is the ONE thing Chemistry is always doing in every single one of these industries? Write your answer in your notebook. You have 2 minutes — no talking.' Enforce 2 minutes of silent writing. Then say: 'Who wants to take a risk and share their answer?' Wait 12 seconds. Cold-call 3 volunteers and 1 non-volunteer. After responses, synthesise: 'Every single one of you is circling the same idea — Chemistry is always about matter changing. That is exactly what happened on the jiko. The chapati is Chemistry at work — just like Kipchoge's muscles, a nuclear reactor, and a firework are all Chemistry at work.' Introduce the CER frame verbally and on the board, then give learners 5 minutes to write individually. Circulate and read 4–5 CER responses silently, noting range of quality for teacher reflection.	Spoke Diagram + Claim-Evidence-Reasoning (CER) Writing. The spoke diagram is a visual synthesis tool that makes the structural relationship between Chemistry and multiple industries simultaneously visible, preventing learners from treating each industry as an isolated topic. The CER frame then internalises the synthesis by requiring each learner to independently articulate a logical chain from evidence to conclusion — the foundational move of scientific explanation. Both strategies anchor back to the phenomenon by design.	Observable indicator: Spoke diagram is populated with chemistry-specific language (e.g., 'nuclear fission releases energy', 'glycogen is a carbohydrate stored in muscles', 'metal salts produce coloured flames', 'sulphur dioxide causes acid rain') — not vague labels. Quality indicator for CER: Teacher scans 4–5 notebooks and checks that (1) the Claim names a specific industry, (2) the Evidence is a named substance or process, and (3) the Reasoning explicitly mentions matter changing or a chemical process. Learners who write 'Chemistry is important' without evidence or reasoning are flagged for targeted support in Phase 4.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Creativity break: How do you apply creativity in	Learners return to the class Driving Question Board displayed at the front. Each learner receives one sticky note. They write ONE new piece of evidence they gathered today that helps answer the driving question: 'How does Chemistry explain what happens	Hand each learner a sticky note and say: 'On one side, write one piece of evidence from today that helps us answer our driving question. On the other side, write one new question today's research has raised for you. You have 3 minutes.' After posting, read 5–6	Sticky-Note Evidence Board with Prediction Revisit. The DQB makes learning progress visible and cumulative across the 10-lesson sequence. The prediction revisit explicitly models the science practice of 'Using Mathematics and Computational Thinking' and	Observable indicator: Every learner posts one evidence note and one new question note. Quality indicator: Evidence notes name a specific chemical substance or process (not 'Chemistry is used in sports'). New question notes are genuinely

[algebra](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?' Learners also write a NEW question that today's research has raised for them. Learners post evidence notes in the 'Evidence We Have' column and question notes in the 'New Questions' column. The teacher reads 5–6 notes aloud, narrating how the evidence gathered in Lessons 1–5 is building a cumulative answer. The teacher highlights that the Prediction Parking Lot predictions from Phase 1 can now be revisited — learners verbally confirm which predictions were accurate, which were partially correct, and which were wrong, and why.	notes aloud, pausing after each: 'This learner says nuclear fission is a chemical change that releases energy — how does that connect to our chapati?' Wait 10 seconds. Then return to the Prediction Parking Lot: 'Look at our predictions from the start of class. Thumbs up if your prediction was confirmed by the research. Thumbs sideways if it was partly right. Thumbs down if you were wrong.' After a show of thumbs, cold-call 2 learners with thumbs-sideways: 'What did you predict and what did the evidence actually show?' Affirm the scientific value of being wrong: 'Scientists revise predictions constantly — that is exactly what we are doing here, just like we are doing with our chapati question.'	'Engaging in Argument from Evidence' (NGSS SEP 5 & 7) — learners must justify whether their prediction held up against evidence. This metacognitive loop strengthens epistemological understanding: knowledge is built by testing and revising ideas, not just receiving information.	curious and Chemistry-specific (e.g., 'How does the body break down the carbohydrates in ugali during a marathon?', 'What radiation does Kenya's geothermal energy release?'). Teacher photographs the DQB after the lesson for tracking purposes. Flag any learners who write only vague statements for follow-up at the start of Lesson 6.
Model Building Phase 📹 **VIDEO:** [Creativity break: How do you apply creativity in algebra](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the cumulative 'Chemistry at Work' model begun in Lesson 1 (a large annotated diagram or concept map displayed/stored in the classroom). In Lesson 1 the model began with 'Chemistry = study of matter and its changes' anchored to the chapati. In Lessons 2–4, branches and agricultural/pharmaceutical/food industry roles were added. Today, learners add four new nodes: Nuclear & Energy, Sports & Nutrition, Entertainment, Environmental Chemistry. Each new node must include: (1) the industry name, (2) one specific chemical substance or process, (3) an arrow labelled 'matter changes because…' connecting back to the central 'Chemistry = study of matter and its changes' hub. Pairs draft additions on mini-whiteboards or paper, then a nominated	Direct learners' attention to the cumulative class model: 'Our model has been growing since Lesson 1. Today we add four new branches. With your expert group partner, draft what your branch's node should say on your mini-whiteboard. You have 3 minutes.' After 3 minutes, invite one representative from each expert group to annotate the class model. After each addition, facilitate a quick class evaluation: 'Does this node pass our three Thumbs Criteria? Show me thumbs.' If a node fails a criterion, say: 'What needs to be added or changed to make this pass? Suggest a revision.' Cold-call 2 learners for revision suggestions. After all four nodes are added, step back and say: 'Look at this model. From Lesson 1 to today, what is the ONE idea at the centre of everything?' Wait 12 seconds.	Cumulative Annotated Concept Map (Progressive Model Building). This strategy embodies NGSS SEP 2 (Developing and Using Models). By physically returning to and extending the same model across every lesson, learners experience knowledge as cumulative and interconnected rather than lesson-by-lesson isolated facts. Each addition must articulate a mechanism ('matter changes because…'), preventing the model from becoming a simple list and requiring causal reasoning. The three Thumbs Criteria give learners a peer-assessment scaffold that reinforces Chemistry-specific language use.	Observable indicator: Four new nodes are added to the class model, each with an industry name, a specific chemical substance or process, and a labelled arrow back to the central hub. Quality indicator: The 'matter changes because…' arrow labels include causal chemical language (e.g., 'nuclear fission splits uranium atoms releasing enormous energy', 'glycogen in muscles is broken down chemically to release glucose for energy', 'metal salts undergo combustion releasing energy as coloured light'). Teacher photographs the updated model for the lesson record. If any node's arrow label is non-causal (e.g., 'matter changes because Chemistry'), teacher prompts the class: 'That arrow needs a mechanism — what is the actual chemical process happening?'

	student from each expert group annotates the class model on the board. The class evaluates each addition using the 'Thumbs Criteria': Does it name a specific chemical substance or process? Does it connect to matter changing? Is it relevant to real Kenyan life? The lesson closes with the teacher pointing to the fully updated model and saying: 'Every branch on this model is a different way Chemistry explains what happened on that jiko. Next lesson, we are going to zoom in on one of the most important questions Chemistry helps us answer — what happens when we put chemical substances into the human body.'	Cold-call 3 learners. Close with the bridging statement connecting to Lesson 6: 'Next lesson, we are going to zoom in on one of the most important questions Chemistry helps us answer — what happens when we put chemical substances into the human body — because Chemistry explains both the medicines that heal us and the substances that harm us.'		before closing the lesson.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did every expert group produce a presentation that included at least one named chemical substance or process — or did any group remain at the level of 'Chemistry is used in this industry'? If so, was the Research Briefing Card sufficiently specific, or does it need richer Kenya-relevant content for the next cohort? (2) During the CER writing, how many learners articulated reasoning that explicitly connected to matter changing — the conceptual spine of the unit? Review the 4–5 notebooks scanned during Phase 3 and identify the proportion demonstrating this connection. (3) Were the Jigsaw Listener Sheet connections to the chapati phenomenon plausible and chemistry-specific, or did learners default to surface-level connections ('both involve heat')? If the latter, the bridging prompts during group work need to be more explicit in future lessons. (4) How was gender balance in cold-calling, group roles (Presenter, Recorder), and model annotation? Deliberately track whether female learners were equally represented in high-visibility roles, given the KICD core competency on gender stereotyping in Chemistry careers. (5) Is the cumulative class model becoming too dense to read clearly? If so, consider transferring it to a larger medium (e.g., a roll of chart paper) before Lesson 6, where the drug-and-substance-use content will require its own set of additions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed four Kenya-relevant images connecting Chemistry to nuclear energy (Olkaria geothermal, KPLC grid), sports nutrition (Kenyan marathon runners using energy gels and electrolytes), entertainment (coloured dyes, metal-salt pyrotechnics at Nairobi festivals), and environmental chemistry (water quality in Lake Victoria, air pollution on Mombasa Road). They observed expert group presentations and populated a spoke diagram connecting all four industries to the central idea of Chemistry as the study of matter and its changes.
What did I learn?	Chemistry operates in many industries beyond the laboratory: nuclear energy involves matter changes at the atomic level (fission); sports nutrition uses Chemistry to understand how carbohydrates (like ugali and energy gels) are broken down in the muscles of athletes like Kipchoge; entertainment relies on chemical reactions (metal salt combustion, synthetic dyes) to create colour and

	spectacle; and environmental Chemistry monitors and protects the matter in Kenya's water, soil, and air. In every case, Chemistry is explaining, predicting, or controlling how matter changes — the same process that transformed chapati dough on the jiko.
How does this explain the phenomenon?	The central explanation across all industries is that Chemistry is the science of matter change. Whether a uranium atom splits in a reactor, glycogen breaks down in a runner's muscle, a copper compound burns green in a firework, or sulphur dioxide reacts with water vapour in Nairobi's air — all of these are matter undergoing chemical changes that Chemistry can describe and explain. This is the same type of change (irreversible, involving new substances and energy transfer) that occurred when raw chapati dough was transformed by heat into a firm golden disc. Understanding these changes allows scientists, engineers, athletes, and policymakers to make safer, smarter decisions — which is exactly what the driving question asks us to explore.

LESSON 6 (80 minutes): Careers in Chemistry and Gender Stereotyping	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to research careers in Chemistry, critically examine how gender stereotypes in Kenyan society influence who pursues Chemistry careers, and advocate for equitable access to Chemistry-related professions.
Knowledge	Learners will identify a range of Chemistry careers (e.g., pharmacist, food scientist, agricultural chemist, nuclear chemist, sports scientist, industrial chemist) and explain how Chemistry knowledge underpins each role.
Skills	Learners will use digital and print resources to gather information, organise findings into a structured presentation, and engage in evidence-based debate about gender stereotyping in career choices.
Attitudes	Learners will demonstrate self-efficacy by confidently presenting findings in plenary, and show respect by acknowledging the value of every peer's contribution regardless of gender or background.
Key Inquiry Question	What is Chemistry and why is it important in our daily lives? / How does drug and substance use affect our health and well-being?
Purpose in Storyline	Having established in Lessons 1–5 what Chemistry is, its branches, the states of matter, and how drugs interact with the body chemically, learners now zoom out to ask: 'Who does Chemistry?' This lesson challenges the implicit assumption that the chemist behind the chapati science, the pharmacist who formulates the medication, or the food technologist who designs safer cooking oil is any one gender. By connecting the irreversible transformation of the chapati dough to the real human careers that investigate and apply such changes, learners see Chemistry as an inclusive, life-changing profession open to all — and they update the Driving Question Board (DQB) with career-related questions that will persist into future sub-strands.
Safety Notes	No chemical substances are handled in this lesson. Ensure respectful, inclusive language during the debate; the teacher should pre-establish group norms around gender-sensitive discourse. If learners share personal experiences related to gender discrimination or drug/substance pressure from peers, apply school safeguarding protocols and refer to the school counsellor as appropriate.

B. LESSON OVERVIEW
Learners begin by reconnecting to the chapati phenomenon and reflecting on who they imagine made the chapati and who they imagine studies the chemistry behind it. They then use the ARES platform and print resources to research Chemistry careers across six sectors (food, pharmaceutical, agriculture, manufacturing, sports, nuclear/entertainment). In expert jigsaw groups, they investigate how gender stereotypes in Kenyan society shape career choices and access. Groups present findings in a structured plenary, followed by a structured academic controversy debate: 'Should Chemistry careers be gendered?' The lesson closes with a DQB update, a model revision, and a teacher-led bridge to the next lesson on drug and substance use — a field where Kenyan chemists and pharmacists play a critical protective role.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:**	The teacher displays a large image split into two panels: (Left) a woman in a kanga apron pressing	Display the split image on the ARES platform projector or printed A3 sheet. Say: 'Before we read	Activating Prior Knowledge + Bias Inventory: By externalising mental images before instruction, learners	Observable indicator: Every learner has written at least one name and one career in their

[Prime numbers](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	chapati dough on a jiko in a Nairobi kitchen; (Right) a scientist in a white lab coat examining a sample in a KEBS (Kenya Bureau of Standards) laboratory. Learners individually write answers to three quick-fire questions in their notebooks — (1) 'Who do you picture when you hear the word 'chemist'?' (2) 'Can you name one Kenyan woman and one Kenyan man working in a Chemistry-related field?' (3) 'Is there a Chemistry career you personally would enjoy? Why or why not?' Learners then share predictions with a shoulder partner for 2 minutes before a brief whole-class share-out. Learners record their initial thoughts on sticky notes and post them under two columns on the class wall: 'Chemistry careers I know' and 'Questions I have about Chemistry careers'.	anything or watch anything today, I want to know what is already in your head. Open your notebooks. You have 3 minutes — silent individual writing. Answer these three questions on the board.' [Post 3 questions visibly.] After 3 minutes, say: 'Turn to your shoulder partner. Share your answers. You have 2 minutes.' [Start timer.] After pair share, cold-call 4 learners — deliberately select at least 2 female learners and 2 male learners — and ask: 'What name did you write for a Kenyan chemist? Male or female?' Record names on the board. Count the gender split publicly and say: 'Interesting — let's keep this tally and see if it changes by the end of the lesson.' Distribute sticky notes and direct learners to post on the wall chart. [WAIT TIME: 10–15 seconds after each cold-call before accepting responses.]	surface implicit gender biases about who 'belongs' in Chemistry. Recording a live tally on the board creates a data artefact the class will revisit — making the bias concrete, measurable, and discussable rather than abstract.	notebook within 3 minutes. The teacher scans notebooks during the pair-share. The board tally of named Kenyan chemists (gender split) serves as a baseline data point to be revisited at lesson end. Red flag: if fewer than 2 female chemists are named across the whole class, the teacher notes this and uses it explicitly in Phase 3.
Observe Phase 📹 **VIDEO:** [Prime numbers](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners are organised into six 'Expert Groups' of 4–5, each assigned one Chemistry career sector. Using ARES platform resource cards and/or printed research sheets, each group investigates: (A) Food Chemistry — e.g., the food technologist who tests whether unga (maize flour) from the Rift Valley meets KEBS standards; (B) Pharmaceutical Chemistry — e.g., the pharmacist or medicinal chemist who formulates dosage for malaria drugs distributed in Kisumu; (C) Agricultural Chemistry — e.g., the agrochemist who develops safe fertilisers for tea farmers in Kericho; (D) Industrial/Manufacturing Chemistry — e.g., the industrial chemist at Bamburi Cement or Bidco Oil	Assign groups before the lesson using the ARES platform's group randomiser — ensure gender-mixed groups. Distribute Research Cards. Say: 'Each group is now an expert team. Your job is not just to find what chemists in your sector do — your job is to find who does it, and ask honestly: does gender play a role in who gets into this career in Kenya?' Circulate to each group at least once in the 15 minutes. At each group, ask one probing question: e.g., to Food Chemistry group — 'You know the person pressing the chapati on the jiko — is that person ever called a chemist? Why not?' To Pharmaceutical group — 'If a young woman from Kibera wants to become a medicinal chemist, what barriers might she face that a	Expert Jigsaw + Structured Research Cards: Distributing expertise across groups means every learner becomes the classroom authority on one sector, creating genuine interdependence during the share-out. Structured Research Cards scaffold evidence gathering so learners don't simply copy facts but must identify a real person and name a stereotype — connecting data to the lesson's critical inquiry. This mirrors the NGSS practice of 'Obtaining, Evaluating, and Communicating Information.'	Observable indicator: Each group's Research Card has all four fields completed with specific, Kenya-relevant evidence (not generic global facts) before the share-out begins. The teacher checks for field 3 (named Kenyan/African person) and field 4 (specific stereotype named) as the minimum acceptable standard. Red flag: if a group writes 'women can't be chemists' as a fact rather than a stereotype to be critiqued, the teacher intervenes immediately with a Socratic question: 'Is that a fact or a belief? How would we test that?'

	Refineries; (E) Sports Science Chemistry — e.g., the sports chemist who analyses performance-enhancing substances for Athletics Kenya; (F) Nuclear/Environmental Chemistry — e.g., the environmental chemist monitoring water quality at Lake Victoria. Each group completes a structured Research Card with four fields: (1) What does this chemist do day-to-day? (2) What Chemistry knowledge do they need? (3) Name one real Kenyan or African person in this field. (4) What gender stereotypes might affect who enters this career in Kenya? Groups have 15 minutes for research, then 5 minutes to prepare a 2-minute oral report.	male classmate might not?' [WAIT TIME: 12 seconds after each probe.] At the 10-minute mark, give a 5-minute warning. At 15 minutes, say: 'Finish your notes. Presenter, stand up — you have exactly 2 minutes. Go.'		
Explain Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each Expert Group's presenter delivers a 2-minute report to the class. After all six reports, the teacher facilitates a Structured Academic Controversy (SAC) debate on the proposition: 'Chemistry careers in Kenya should be equally accessible to all genders.' The class is split into two large groups: Team A argues FOR equal access (using evidence from the research), Team B argues the current reality IS that stereotypes exist and presents those stereotypes as evidence of systemic barriers — not as endorsements. After 5 minutes of argument from each side, teams swap positions and argue the opposite view for 3 minutes each. The debate closes with a 3-minute open synthesis discussion where the whole class together crafts a single consensus statement. Learners then return to the board tally from Phase 1 and update it — adding any new Kenyan chemists	After each 2-minute group presentation, ask one follow-up question to the whole class — cold-call a learner NOT in the presenting group. E.g., after Food Chemistry report: 'Wanjiku, your group researched pharmaceuticals — how does what the food chemist does connect to what a medicinal chemist does? Think about the chapati change — who studies the chemistry of what we eat and who studies the chemistry of what we take as medicine?' [WAIT TIME: 15 seconds.] Before the SAC debate, establish clear norms: 'In this debate, we are arguing positions, not attacking people. Critique the idea, never the person.' Assign Team A and Team B by row. After the swap, watch for learners who struggle to argue the opposite position — this is a high-value teachable moment. During synthesis, ask: 'What single statement can we ALL agree on about Chemistry careers and	Structured Academic Controversy (SAC): SAC is a cooperative learning structure where learners must argue both sides of a proposition before reaching consensus. This builds intellectual flexibility, evidence-based reasoning, and the ability to distinguish personal opinion from supported argument — directly developing the Self-Efficacy and Communication and Collaboration core competencies mandated by KICD. The position-swap is critical: it prevents entrenchment and models the scientific habit of considering alternative explanations. Connecting the final consensus statement back to the chapati phenomenon (who studies it, who benefits from that study) anchors the social justice discussion in the unit's scientific core.	Observable indicator: Each learner's 3-sentence Chemistry Career Statement in their notebook contains (1) a clear personal position, (2) a piece of evidence from today's research, and (3) a connection to the real world (Kenyan context). The teacher reads 4–5 statements aloud (with permission) as a class check. Red flag: statements that are purely opinion with no evidence signal that the research phase did not produce usable knowledge — the teacher notes these learners for targeted follow-up.

	discovered during research. The updated tally is photographed and stored in the class ARES project folder. Learners individually write a 3-sentence 'Chemistry Career Statement' in their notebooks: 'I believe… because… and the evidence from today's research shows…'	gender in Kenya?' Write the consensus statement live on the board. Return to the board tally: 'We started with [X] named chemists — [Y] female, [Z] male. What does our research add?' Cold-call 3 learners to add names. Say: 'This tally is evidence. Scientists use evidence. What does this evidence tell us?' [WAIT TIME: 12 seconds.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class DQB (established in Lesson 1 and updated each lesson). The teacher reads aloud the Driving Question: 'How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?' Learners individually write one new question and one new evidence statement on separate sticky notes. New question examples might include: 'Who are the chemists in Kenya making sure our food, medicines, and water are safe?' 'Why are there fewer women in Chemistry degree programmes in Kenyan universities?' 'How does knowing Chemistry help a sports coach at Athletics Kenya make smarter choices?' Evidence statement examples: 'I now know that an agricultural chemist in Kericho uses Chemistry to make tea farming safer.' 'Gender stereotypes in Kenya reduce the number of girls who pursue Chemistry careers.' Learners place BLUE sticky notes for new questions and GREEN sticky notes for new evidence on the DQB. The teacher then draws the class's attention to how the DQB has grown since Lesson 1 —	Say: 'Take two sticky notes — one blue, one green. Blue is a new question today's lesson raised for you. Green is a piece of evidence you are now confident about.' Give 3 minutes silent writing. Then: 'Stand up, walk to the DQB, post your notes, and read two notes already on the board that you find interesting.' Allow 3 minutes of movement. Once learners are seated, cold-call 3 learners: 'Ochieng, you read an interesting note — which one and why?' [WAIT TIME: 10 seconds.] Then stand at the DQB and physically trace the journey: 'In Lesson 1, we asked what Chemistry is. In Lesson 2, we explored its branches. In Lessons 3–4, we looked at matter and its states. In Lesson 5, we explored why the chapati change is chemical. Today we asked who does Chemistry — and we found evidence of real Kenyan scientists in every sector. Our DQB shows us that our understanding is growing, lesson by lesson, like layers of chapati dough building into something new.' [Record the total count of DQB notes: questions vs evidence — this is a formative data point for the teacher.]	Driving Question Board (DQB) Curation: The DQB functions as a living epistemic map — a public record of what the class knows and still wonders about. By colour-coding questions (BLUE) versus evidence (GREEN), learners visually track the ratio of uncertainty to certainty, which is a metacognitive scaffold. The teacher's verbal trace of the lesson arc (Lessons 1–6) reinforces narrative coherence in the unit storyline and helps learners see their own intellectual progress — a core driver of self-efficacy.	Observable indicator: Every learner posts at least one blue and one green sticky note. The teacher scans for green notes that are vague ('Chemistry is important') versus specific ('An agricultural chemist in Kericho tests fertiliser safety') — specificity of evidence statements is the quality indicator. The teacher photographs the DQB after the lesson and adds it to the ARES class digital portfolio.

	connecting the career questions to the earlier questions about what Chemistry is and what it explains.			
Model Building Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the personal Chemistry Concept Map they began in Lesson 1 (saved in their ARES learner portfolio or drawn in their notebooks). In Lessons 1–5, the map included: Chemistry definition → branches → matter/states of matter → physical vs. chemical changes → the chapati phenomenon. Today, learners add a new node: 'Careers in Chemistry' with at least six labelled branches (one per sector studied), and a second new node: 'Who does Chemistry?' with connections to gender equity, Kenyan scientists named today, and the statement 'Chemistry is for everyone.' Learners also draw an arrow from 'Careers in Chemistry' back to 'Chapati Phenomenon' and label the arrow: 'Real chemists study and apply reactions like this every day.' Finally, the teacher gives a 2-minute bridge preview: 'In our next lesson, Lesson 7, we will investigate drug and substance use — this is a field where pharmacists and medicinal chemists — including the Kenyan scientists we named today — play a critical role. The same Chemistry that explains the chapati change also explains how a drug changes your body's chemistry. We need to understand that science so we can make safer, smarter choices.' Learners write the bridge question in their notebooks: 'How does a chemist's knowledge protect us from the dangers of drug and substance use?'	Say: 'Open your concept maps. Today we grew our understanding of who Chemistry belongs to. Add the two new nodes I have written on the board.' [Write nodes and branch labels on the board clearly.] Circulate for 5 minutes, checking that the arrow from 'Careers in Chemistry' to 'Chapati Phenomenon' is present and labelled. Prompt any learner who has not drawn it: 'Who made the chapati on the jiko? Who studied the chemistry behind it? Connect those two nodes and tell me why.' [WAIT TIME: 12 seconds.] Cold-call 2 learners to share their completed connection aloud. Close with the bridge preview — deliver it with deliberate pace and energy: 'Next lesson, same chapati. But now we ask — what happens when we put a chemical substance — a drug — into a human body? And who are the chemists responsible for keeping that safe?' Pause. 'Write that question down. That is your homework question to think about tonight.' Dismiss class.	Cumulative Concept Map Revision (Model-Based Reasoning): Requiring learners to physically add to a map built over six lessons makes conceptual growth visible and tangible. The explicit arrow from 'Careers in Chemistry' back to the 'Chapati Phenomenon' prevents the careers lesson from feeling like a detour — it re-anchors the social justice content in the unit's scientific core. The bridge question activates anticipatory curiosity for Lesson 7, a technique that improves knowledge retention and lesson-to-lesson coherence. This aligns with the NGSS practice of 'Developing and Using Models.'	Observable indicator: Learner concept maps have (1) at least six career branches correctly labelled with Kenyan-relevant sector names, (2) the 'Who does Chemistry?' node with at least two connections, and (3) a labelled arrow to the chapati phenomenon. The teacher collects 5 maps (random sampling) and reviews them after class using the ARES portfolio tool. The bridge question written in the notebook is checked in Lesson 7's opening warm-up as a prior-knowledge activation check.

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions before Lesson 7: (1) Evidence of self-efficacy — Did learners who are typically quiet or hesitant volunteer to present during the plenary? If not, what scaffolding (sentence starters, pre-assigned roles, pair rehearsal) could reduce the barrier to participation? (2) Quality of gender discourse — Did the SAC debate remain evidence-based, or did it drift into personal anecdote or stereotyping? If stereotypes were stated as facts rather than critiqued, revisit norms at the start of Lesson 7. (3) DQB growth — Count the ratio of BLUE (question) to GREEN (evidence) notes. If questions vastly outnumber evidence, learners may still be in early sensemaking and need more structured evidence-gathering before Lesson 7's drug/substance discussion. (4) Concept map depth — Do learner maps show genuine integration (arrows between nodes, labels on arrows) or are they lists? Shallow maps suggest learners are cataloguing information rather than building relational understanding — plan a brief map-reading protocol at the start of Lesson 7. (5) Kenyan scientist representation — Were female Kenyan chemists named in enough numbers to shift the Phase 1 tally significantly? If not, prepare a short ARES resource card for Lesson 7 featuring 3–4 named Kenyan women in Chemistry (e.g., researchers at the University of Nairobi, KEMRI, KEBS, or KIRDI) to ensure representation is not left to chance.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed (via ARES platform and print research cards) that Chemistry careers span six major sectors — food, pharmaceutical, agricultural, industrial, sports, and nuclear/environmental — and that real Kenyan scientists work in every one of these sectors. They also observed, through their own Phase 1 board tally, that the names they initially associate with 'chemist' are disproportionately male, and that this disparity is a product of social stereotype rather than scientific aptitude.
What did I learn?	Learners learned that Chemistry underpins careers that directly affect everyday Kenyan life — from the KEBS food technologist certifying safe unga, to the KEMRI medicinal chemist formulating malaria drugs, to the agrochemist protecting Kericho tea farms. They learned that gender stereotypes in Kenyan society create systemic barriers that reduce equitable access to Chemistry careers, and that self-efficacy — the belief in one's own ability to succeed — is a critical tool for overcoming those barriers. They also consolidated prior learning by connecting each career sector back to the core idea that Chemistry is the study of matter and its changes — including the irreversible change that turns chapati dough into a golden disc.
How does this explain the phenomenon?	Learners can now explain, using evidence from their jigsaw research, that Chemistry careers are not defined by gender but by knowledge, skill, and interest — and that the same chemical principles that explain the chapati phenomenon (matter transformation, energy, irreversible change) are applied daily by Kenyan chemists across industries that feed, heal, protect, and empower Kenyan communities.

LESSON 7 (80 minutes): Meaning of Drug, Prescription, Dosage and Substance Use	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To develop learners' understanding of key pharmaceutical terminology — drug, prescription, dosage, and substance use — and to build their capacity as informed, responsible consumers of medicines and chemical substances.
Knowledge	Learners explain the meaning of drug, prescription, dosage, over-the-counter (OTC) medicine, prescription-only medicine, substance use, and substance abuse; and identify how dosage information appears on real medicine labels (Panadol, Metformin).
Skills	Learners read and interpret medicine labels for dosage, caution, and expiry information; distinguish between appropriate substance use and misuse; and construct reasoned arguments about responsible medicine use using evidence from labels.
Attitudes	Learners demonstrate critical and constructive dialogue while brainstorming with peers on consumer rights to information on drug prescription, substance use, and knowledge of side effects for awareness (Citizenship competency — verbatim KICD).
Key Inquiry Question	How does drug and substance use affect our health and well-being?
Purpose in Storyline	In previous lessons learners established that Chemistry studies matter and its changes — including the irreversible chemical change that turns chapati dough into a cooked chapati. In Lesson 7 the storyline takes a pivotal turn: learners now examine how the pharmaceutical branch of Chemistry produces drugs — chemical substances designed to cause deliberate, controlled changes in the human body. Just as exceeding heat on the jiko ruins the chapati irreversibly, exceeding a prescribed dosage or misusing a substance can cause irreversible harm to the body. This lesson equips learners to read the chemical 'instructions' (labels/prescriptions) that chemists and doctors use to keep those changes safe and beneficial.
Safety Notes	No hazardous chemicals are used in this lesson. Medicine labels/packages used are for reading and analysis only — no medicines are consumed. Remind learners that medicines should only be taken under adult supervision and as directed. If any learner discloses personal experience with substance abuse during discussion, respond with empathy, avoid stigmatising language, and follow the school's safeguarding protocol.

B. LESSON OVERVIEW

Learners read a structured 'Science of Drugs' article set in a Kenyan context, then analyse printed Panadol and Metformin medicine labels in pairs. Through guided discussion and a Frayer Model vocabulary activity, they define drug, prescription, dosage, OTC medicine, prescription-only medicine, substance use, and substance abuse. A Socratic whole-class discussion connects responsible medicine use to consumer rights (PCI: Consumer Protection; PCI: Drug and Substance Use). The lesson closes with an update to the class Driving Question Board (DQB), where learners record new evidence linking Chemistry — specifically the pharmaceutical industry — to the anchoring phenomenon of irreversible chemical change in the body. Citizenship is the foregrounded core competency.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown two printed images on the projector or board:	Display both images simultaneously. Say verbatim:	Think-Pair-Share with Prediction Parking Lot. This activates prior	Teacher scans written predictions during circulation — look for

📹 **VIDEO:** [Human activities that threaten biodiversity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Discovering Medicine..:Handout Chemical Structures] Source: MIT Blossoms 🔍 Search ARES for similar readings	(A) a ball of raw chapati dough on a jiko, and (B) a strip of Panadol tablets beside a handwritten doctor's prescription slip. The teacher poses a Think-Pair-Share prompt: 'Both the jiko and this Panadol tablet cause a change inside something — the dough and your body. Before we read anything today, predict: what do you think the word DOSAGE means, and why might getting it wrong be dangerous?' Learners write their prediction individually for 2 minutes, then share with a shoulder partner for 2 minutes. Four pairs are cold-called to share. Learners record predictions in their science notebooks under the heading 'My Prediction — Lesson 7.'	'Look at these two images. In Lesson 1 we saw that too much heat on the jiko caused an irreversible change to our chapati. Today we are asking — what happens when a chemical substance causes a change inside your body, and how do chemists and doctors control that change?' Pose the Think-Pair-Share prompt. Allow 2 minutes of silent individual writing — enforce silence, circulate, do NOT answer questions. After pair discussion, cold-call exactly 4 pairs using a random name stick. For each response, probe with: 'What clue in everyday life gave you that idea?' Allow 10–12 seconds WAIT TIME after each cold-call before accepting answers. Record student-generated prediction words on the board in a 'Prediction Parking Lot' column — these will be revisited in the Explain phase.	knowledge about chemical change (from Lessons 1–6) and creates cognitive dissonance — learners will later discover whether their intuitive definition of 'dosage' matches the precise scientific/pharmaceutical meaning. The visual link back to the chapati phenomenon keeps the anchoring event alive and signals that today's content is not disconnected from previous learning.	whether learners connect 'dosage' to quantity/amount (correct intuition) vs. vague ideas like 'type of medicine' (misconception to address). Cold-call responses reveal whether learners understand that chemical substances cause changes in the body — a prerequisite concept for this lesson. Indicator of readiness: at least 3 of 4 cold-called pairs mention quantity, amount, or how much in their prediction.
Observe Phase 📹 **VIDEO:** [Human activities that threaten biodiversity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Discovering Medicine..:Handout Chemical Structures] Source: MIT Blossoms 🔍 Search ARES for similar readings	PART A — Article Reading (12–25 min): Learners receive a one-page 'Science of Drugs' article written at Grade 10 reading level. The article features a fictional Kenyan pharmacist, Amina Ochieng of Kisumu County Referral Hospital, explaining to a patient the difference between OTC drugs (e.g., Panadol for fever) and prescription drugs (e.g., Metformin for diabetes). The article defines: drug, prescription, dosage, OTC medicine, prescription-only medicine, substance use, and substance abuse, with each term bolded. Learners read individually and highlight (or underline if no highlighters) each bolded term and its definition. They then complete a half-completed Frayer Model template in their notebooks for the	Distribute the article and say: 'Read silently and highlight every bolded term and its definition. You have 8 minutes — GO.' Set a visible timer. At the 8-minute mark, say: 'Now complete the Frayer Model for the word DRUG only — use the article as your source. You have 4 minutes.' After Frayer Models are completed, do a rapid gallery check — circulate and place a tick or a question mark on 5–6 notebooks to flag correct vs. incomplete entries. Transition to Part B: 'Now you are going to be Label Detectives. Every medicine label is a chemistry document — it tells you exactly what chemical is in the tablet, how much, and how often. With your partner, complete the Label Detective sheet for BOTH labels.' Circulate during pair	Frayer Model Vocabulary Strategy + Label Detective Structured Observation. The Frayer Model forces learners to move beyond a simple definition to distinguish examples from non-examples — a critical thinking skill required for understanding the drug vs. non-drug distinction. The Label Detective sheet scaffolds scientific reading of real pharmaceutical documents, mirroring what chemists and pharmacists do professionally, and directly addresses the KICD PCI on Consumer Protection (reading product labels for quality marks, expiry dates, cautions).	Collect or photograph 5 Frayer Models during circulation — check that: (a) Definition matches the article, not a vague colloquial meaning; (b) Kenya-relevant examples are listed (Panadol, Metformin, Septrin, ORS sachets); (c) Non-examples are genuinely not drugs (e.g., ugali, water — though teacher should probe water as a borderline case). On the Label Detective sheet, the critical observable indicator is whether pairs correctly identify that Metformin requires a prescription while Panadol does not, AND can quote the label text as evidence.

	term 'DRUG' (Definition / Characteristics / Examples from Kenya / Non-examples). PART B — Label Analysis (25–35 min): In pairs, learners receive printed label inserts for Panadol (500 mg tablets) and Metformin (500 mg tablets). They use a structured 'Label Detective' observation sheet to record: (1) Name of drug, (2) Active ingredient and amount, (3) Recommended dosage for adults, (4) Maximum dosage in 24 hours, (5) Caution/warning statements, (6) Expiry date location, (7) Whether a prescription is required. Pairs compare findings for both labels.	work. Stop the class at the 3-minute warning: 'One minute left — make sure you have answered question 7: does this medicine need a prescription? Be ready to justify your answer.' Cold-call 3 pairs (one per question cluster) to share Label Detective findings. After each pair shares, ask the class: 'Does anyone have a different answer? Thumbs up if you agree, thumbs sideways if you found something different.' Allow 10 seconds WAIT TIME before resolving discrepancies.		
Explain Phase 📹 **VIDEO:** Human activities that threaten biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Discovering Medicine..:Handout Chemical Structures Source: MIT Blossoms 🔍 Search ARES for similar readings	PART A — Vocabulary Consolidation (35–43 min): Using the completed Frayer Models and Label Detective sheets, the class co-constructs a 'Class Definition Wall' — a large chart paper on the board with seven boxes, one per key term. Each group of 4 learners is assigned one term and given 3 minutes to write a class-agreed definition (in their own words, not copied from the article) plus one Kenya-relevant example and one consequence of misuse. Groups post their term card and a spokesperson reads it aloud. Class votes on clarity using a thumbs up/sideways/down system. Teacher refines language where needed. PART B — Socratic Discussion: Chapati-to-Body Connection (43–55 min): Teacher re-displays the chapati phenomenon image and asks: 'In Lesson 1 we said the chapati change is irreversible — you cannot un-bake it. Using what you now know about drugs and dosage, can someone explain	For Part A: Assign groups by counting off 1–7. Give each group a half-sheet of chart paper and a marker. Say: 'Do NOT copy from the article. Write the definition in words a Form 1 student who missed today's lesson would understand. You have exactly 3 minutes.' Time strictly. During the gallery walk, stand back — allow learners to read silently for 60 seconds before opening voting. When a definition is vague, say: 'I hear what you mean — can you add an example from Kenya to make it concrete?' For Part B Socratic Discussion: After groups discuss, cold-call 5 learners using name sticks. For the first response, say nothing — wait 12 seconds, then ask: 'Who wants to build on that idea?' Use the phrase 'What is your EVIDENCE from the label or the article?' at least twice. When a learner makes the chapati–body parallel (irreversibility), explicitly affirm: 'That is exactly the kind of scientific thinking Chemistry develops — connecting a kitchen	Co-constructed Definition Wall + Socratic Questioning with Phenomenon Reconnection. The Definition Wall makes vocabulary learning a collaborative, public, and negotiated process — learners must justify their wording to peers, deepening retention. The Socratic reconnection to the chapati phenomenon is the critical sensemaking move: it anchors the abstract concept of dosage and substance abuse to the concrete, memorable image of irreversible chemical change that opened the unit, making the pharmaceutical content feel like a natural extension of chemistry rather than a separate topic.	Observe group Definition Wall entries for accuracy and Kenya-relevance. Listen for the key conceptual leap during Socratic discussion: do learners spontaneously use the word 'irreversible' to describe the effects of substance abuse? This is the target understanding. Specific indicator: at least 3 of the 5 cold-called learners make a logical connection between exceeding a prescribed dosage and causing uncontrolled/harmful chemical changes in the body — mirroring the uncontrolled heat that irreversibly changes the chapati.

	what happens when a person takes too much of a drug — or takes a drug not prescribed for them? Is that change also irreversible?' Learners discuss in groups of 4 for 3 minutes, then share in plenary. Teacher introduces the concept of substance abuse and its irreversible effects on the body (liver damage, addiction, organ failure) using 3 bullet points on the board — framed as: 'Chemistry explains WHY these changes happen at the molecular level.' PART C — Rights and Responsibilities (55–58 min): Teacher briefly highlights learner rights and responsibilities to a safe and healthy learning environment — verbatim PCI: Life Skills (safety and security). Pose: 'As a Chemistry student, what is ONE responsibility you have toward your own body as a chemical system?'	phenomenon to molecular changes in the human body.' Write the word IRREVERSIBLE on the board next to both the chapati image and a body icon. For Part C: Pose the question, allow 10 seconds of silent thinking, take 3 hands (do not cold-call for this sensitive reflection), and close with: 'Your body is a chemical system. Chemistry gives you the knowledge to protect it.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Human activities that threaten biodiversity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Discovering Medicine..:Handout Chemical Structures] Source: MIT Blossoms 🔍 Search ARES for similar	Learners return to the class Driving Question Board (DQB) displayed at the front of the room. The teacher reads aloud the main driving question: 'How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?' Each learner receives two sticky notes: (1) a GREEN note — 'New evidence I gathered today that helps answer the driving question'; (2) an ORANGE note — 'A new question today's lesson raised for me.' Learners write individually for 3 minutes and post their notes in the designated columns on the DQB. The teacher reads 4–5 green notes aloud and maps them	Hand out sticky notes before learners leave their seats. Say: 'Green note: write one specific piece of evidence — something you can point to on a label, in the article, or from our discussion — that helps answer our big driving question. Orange note: write a genuine question you still have. You have 3 minutes — silence please.' Enforce the silent writing period. As learners post notes, circulate and read them. Select 4–5 green notes that represent the range of evidence quality (one strong, one partial, one needing refinement). Read each aloud and say: 'This evidence is strong because…' or 'This is a good start — what label evidence would make it even stronger?' Point to	Driving Question Board with Colour-Coded Evidence Tracking. The two-colour sticky-note system separates EVIDENCE (green — answers) from INQUIRY (orange — new questions), modelling how scientists distinguish between what is known and what remains to be investigated. Publicly mapping green notes to the driving question makes the cumulative nature of evidence-building visible and motivates learners to contribute quality evidence rather than vague statements.	Read all green sticky notes after class. Sort into three categories: (A) Strong — cites specific label evidence or article content and connects to Chemistry; (B) Partial — makes a connection but without specific evidence; (C) Weak/off-topic — vague or unrelated. Target: ≥ 70% of green notes fall into Category A or B. Any learner whose note falls into Category C should be followed up with a targeted question at the start of Lesson 8.

readings	to the evidence tracker at the bottom of the DQB, adding: 'Drugs are chemical substances — Chemistry explains how they work AND how to use them safely.' The teacher also reads 2–3 orange notes and circles any questions that will be addressed in Lessons 8–10.	the DQB evidence tracker and add: 'Today we confirmed that Chemistry is inside every medicine cabinet in Kenya. The pharmaceutical industry uses Chemistry to make controlled, safe changes to the human body — the same science that explains our chapati.' Circle 2 orange questions that preview Lessons 8–10 (e.g., questions about addiction, body systems, or drug testing).		
Model Building Phase 📹 **VIDEO:** Human activities that threaten biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Discovering Medicine..:Handout Chemical Structures Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners open to their cumulative 'Chemistry Explains Our World' model page in their notebooks — a running annotated diagram they have been building since Lesson 1. The model currently shows the chapati transformation with labels for physical and chemical change, branches of Chemistry, and career pathways. Today, learners add a new section: 'THE PHARMACEUTICAL BRANCH — Chemistry in the Body.' They draw a simple outline of the human body and annotate it with: (1) An arrow labelled 'DRUG (chemical substance) enters body'; (2) A box labelled 'DOSAGE — the controlled amount that causes a safe, intended change'; (3) A red warning box labelled 'SUBSTANCE ABUSE — uncontrolled chemical change → irreversible harm'; (4) A link arrow connecting their body diagram back to the chapati image already on the model, labelled 'Both involve irreversible chemical change — Chemistry helps us control it.' Learners share their updated model with a partner for 2 minutes of peer feedback using the prompt: 'One thing my partner's model explains clearly / One thing that could be clearer.'	Project a partially completed model template on the board showing the body outline and the three annotation boxes as scaffolds. Say: 'Open your model page. You are going to add the pharmaceutical branch today. Use the template on the board as a guide — but your annotations must be in your OWN words, not copied from the board.' Allow 7 minutes of individual model work. Circulate and look for the critical link arrow — learners who do not draw the connection back to the chapati phenomenon should be prompted: 'How does this connect to where our story started — the jiko and the dough?' After 7 minutes, say: 'Turn to your model partner. You have 2 minutes. Tell them one thing their model explains clearly and one thing you think could be made clearer. GO.' After peer feedback, cold-call 2 pairs to share one insight from their peer review. Close by saying: 'Each lesson, your model grows. By Lesson 10, it will tell the complete story of Chemistry — from a ball of dough on a jiko to the molecules inside every medicine cabinet in Kenya.'	Cumulative Annotated Model Revision with Peer Feedback. The running model is the primary sensemaking artefact of the unit — it makes thinking visible, cumulative, and revisable, mirroring how scientific models are refined as new evidence is gathered. The mandatory link arrow to the chapati phenomenon is the structural device that prevents learners from treating pharmaceutical chemistry as an isolated topic, reinforcing the unit's conceptual coherence. Peer feedback using a structured prompt (clear / could be clearer) develops scientific communication skills without requiring formal assessment language.	Circulate during model building and check for three specific elements: (1) The 'DOSAGE' annotation correctly describes a controlled, intentional amount — not just 'how much medicine'; (2) The 'SUBSTANCE ABUSE' box includes the word 'irreversible' or an equivalent phrase; (3) The link arrow to the chapati is present and labelled. Collect 6 notebooks at the end of class (2 strong, 2 average, 2 weak based on in-class observation) to review model quality and inform differentiation in Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. CONCEPTUAL TRANSFER — Did learners spontaneously use the word 'irreversible' (or equivalent) when discussing substance abuse? If fewer than half the class made this connection during the Socratic discussion, plan a brief 5-minute recap at the start of Lesson 8 using the chapati image and a side-by-side comparison of 'controlled heat → good chapati' and 'controlled dosage → safe medicine effect.'
2. VOCABULARY DEPTH — Review the 6 collected notebooks. Are learners writing definitions in their own words, or copying verbatim from the article? If copying is prevalent, introduce a 'Close the Article' routine in Lesson 8 where learners must define terms with the article face-down.
3. CITIZENSHIP COMPETENCY — Did the Socratic discussion remain constructive and non-stigmatising when substance abuse was discussed? If any learner made dismissive or stigmatising comments, use the start of Lesson 8 to revisit the school's values of Respect (KICD verbatim) — 'The learner shows acceptance while appreciating the opinion of each member.'
4. LABEL READING SKILL — Were learners able to locate dosage and caution information independently on the Metformin label, which uses more technical pharmaceutical language than the Panadol label? If pairs struggled, source a simpler OTC label (e.g., ORS sachets widely available in Kenya) for additional practice in Lesson 8.
5. DQB QUALITY — Review the green sticky notes. Are learners citing SPECIFIC evidence (label text, article quotes) or making vague general claims? If fewer than 70% of notes are in Category A/B, model a 'strong evidence statement' at the start of Lesson 8 using a think-aloud: 'I know Chemistry is connected to medicine because the Metformin label says…'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners read a 'Science of Drugs' article featuring Kenyan pharmacist Amina Ochieng of Kisumu County Referral Hospital, and analysed printed Panadol and Metformin medicine labels using a structured 'Label Detective' observation sheet, recording active ingredients, dosage, caution statements, expiry dates, and prescription requirements.
What did I learn?	Learners defined drug, prescription, dosage, over-the-counter (OTC) medicine, prescription-only medicine, substance use, and substance abuse; identified that Metformin requires a prescription while Panadol does not; and co-constructed a Class Definition Wall of the seven key terms with Kenya-relevant examples.
How does this explain the phenomenon?	Learners connected the pharmaceutical branch of Chemistry to the anchoring phenomenon — just as uncontrolled heat causes irreversible changes to chapati dough on the jiko, uncontrolled or excessive intake of chemical substances (drugs) causes irreversible harm to the human body; understanding Chemistry — specifically dosage and prescription information — empowers Kenyans to make safer, smarter choices about their health.

LESSON 8 (80 minutes): Effects of Drug and Substance Use: Tobacco — Chemistry of an Irreversible Harm	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners examine the chemical basis of tobacco's harmful effects, argue from evidence why 'natural' does not mean 'safe', and connect irreversible chemical changes in the body to the anchoring phenomenon of chapati dough on the jiko.
Knowledge	Learners will know that tobacco smoke contains harmful chemicals — tar, nicotine, and carbon monoxide — each of which causes specific, largely irreversible damage to body tissues and systems at the molecular and cellular level.
Skills	Learners will be able to: (i) analyse video and text evidence to identify chemical mechanisms of tobacco harm; (ii) construct a written argument from evidence; (iii) revise their cumulative model to show how chemistry explains irreversible changes in living matter.
Attitudes	Learners advocate for a safe and healthy learning environment by demonstrating critical, evidence-based reasoning about drug and substance use rather than relying on cultural myths or peer pressure.
Key Inquiry Question	How does drug and substance use affect our health and well-being?
Purpose in Storyline	This is Lesson 8 of 10 in the evidence-gathering sequence. Learners have previously established what Chemistry is, its branches, its role in daily life, and the meaning of drugs, prescriptions, and dosage. Now they apply their chemical understanding to tobacco — the most widely abused legal substance in Kenya — deepening SLO (c): examine the effects of drug and substance use in day-to-day life, and SLO (d): advocate for a safe and healthy learning environment. The tobacco focus advances the driving question by showing that Chemistry explains not just changes in food matter (chapati) but irreversible changes in living matter (lung tissue, haemoglobin, nerve cells), reinforcing the unit's central argument that chemical knowledge enables safer, smarter choices.
Safety Notes	No tobacco products are brought into the classroom. If any learner has a family member affected by tobacco-related illness, the teacher acknowledges this sensitively before the video and reminds learners that the discussion is scientific and non-judgmental. Learners showing distress are given permission to step out briefly. The teacher must NOT share personal opinions on tobacco users — frame all discussion around chemical evidence only.

B. LESSON OVERVIEW
Learners open by revisiting the chapati phenomenon and the Driving Question Board to surface what they already know about irreversible chemical changes. They then make a prediction: 'Is tobacco harmful because it is man-made, or could a natural plant also be harmful?' After recording predictions, they watch a focused video on tobacco's three key chemicals (tar, nicotine, carbon monoxide) and annotate a simple body-system diagram. In the Explain phase, groups construct an evidence-based argument responding to the provocation 'The tobacco plant is natural — so why is smoking harmful?' using a claim–evidence–reasoning (CER) frame. They update the DQB and close by revising their cumulative unit model to show that Chemistry explains irreversible changes in both food matter (chapati dough) and living matter (lung tissue). The lesson integrates PCIs on health promotion and life skills, and develops the core competencies of Communication and Collaboration and Citizenship.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners individually read the	Teacher writes the provocation on	Strategy: Anticipatory Prediction	Observable indicator: Every

📹 **VIDEO:** [Jamestown - the impact of tobacco] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	provocation statement written on the board: 'A tobacco plant grows naturally in the soil — just like the wheat used to make chapati dough. So smoking tobacco should be as safe as eating chapati.' Each learner silently writes their prediction (Agree / Disagree / Unsure) and one sentence of reasoning in their science notebooks. Learners then do a 'silent gallery' — they hold up their notebooks so peers can see their stance without verbal discussion. After the gallery, the teacher cold-calls 4 learners (one from each stance category) to read their reasoning aloud. Learners do not change their written prediction yet — it will be revisited at the end of the lesson.	the board in large text and reads it aloud twice. Say exactly: 'Before we watch anything or read anything, I want YOUR thinking — not a correct answer. Write your prediction and one reason. You have 3 minutes — start now.' [Allow 3 minutes of silent individual writing.] After the silent gallery display, say: 'I am going to cold-call four people — one who agrees, one who disagrees, one who is unsure, and one who changed their mind just from looking at classmates.' Cold-call by name (not volunteers). After each response, say: 'Thank you — hold that thought.' Do NOT affirm or correct. After all four, say: 'Keep your prediction visible on your desk — at the end of this lesson you will decide whether the chemistry changes your mind.' [WAIT TIME: 10 seconds after each cold-call before moving to the next learner.]	with Silent Gallery. Learners externalise prior knowledge before instruction, making their thinking visible to themselves and the teacher. The silent gallery creates low-stakes peer comparison that surfaces the range of intuitions in the room (many will assume 'natural = safe') without creating social pressure to conform. This activates the cognitive conflict that the video evidence will then resolve.	learner has a written prediction and at least one sentence of reasoning in their notebook before the video begins. Teacher scans the room during the 3-minute writing period and notes (mentally or on a clipboard) the distribution of Agree / Disagree / Unsure — this informs the depth of scaffolding needed in Phase 3. Red flag: If more than half the class agrees with the provocation ('natural = safe'), the teacher plans to pause the video after the nicotine segment to insert a bridging question.
Observe Phase 📹 **VIDEO:** [Jamestown - the impact of tobacco] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners receive a pre-printed 'Tobacco Evidence Organiser' — a simple outline of the human body (head, chest, blood vessels) with three labelled boxes alongside it: TAR, NICOTINE, CARBON MONOXIDE. Each box has two sub-fields: 'What it is' and 'What it does to the body.' The teacher plays a focused video (8–10 minutes) on the effects of tobacco smoke. During the video, learners annotate their organiser individually. After the video, learners spend 4 minutes comparing annotations with a shoulder partner (Think-Pair) and adding anything they missed. The teacher then cold-calls 3 pairs to share one key fact each about a different chemical. Learners circle	Before playing the video, distribute the Tobacco Evidence Organiser and say: 'Your job during this video is to be a chemistry detective. For each of the three chemicals — tar, nicotine, and carbon monoxide — write what it is and what it does inside the body. Use the organiser — do not just watch.' [Play video. Pause after the tar segment — approximately 3 minutes in — and ask: 'What have you written for tar so far? Thumb up if you have at least one idea.' Resume.] After the video, say: 'Compare with your partner for 4 minutes — add anything you missed in a different colour pen.' After pair-share, cold-call 3 pairs: 'Pair A — tell me one thing you wrote about NICOTINE. Pair B — one thing about TAR.	Strategy: Annotated Evidence Organiser with Red-Pen Irreversibility Coding. The structured organiser prevents passive video watching by binding observation to a specific recording task. The red-pen coding directly links the new evidence (tobacco damage) to the anchoring phenomenon's core concept (irreversibility), so learners begin building the conceptual bridge before the Explain phase. The mid-video pause functions as a comprehension checkpoint and keeps attention active.	Observable indicator: Learners have at least two entries in each of the three chemical boxes by the end of the pair-share. Teacher circulates during pair-share and looks for accuracy of the three mechanisms — specific red flags include: (i) confusing nicotine (addictive stimulant) with tar (carcinogen), (ii) omitting carbon monoxide's displacement of oxygen from haemoglobin. If these errors are common, the teacher writes the three chemicals as column headers on the board and briefly clarifies before Phase 3.

	any effect they describe as 'irreversible' in red pen.	Pair C — one thing about CARBON MONOXIDE.' [WAIT TIME: 12 seconds between cold-calls.] Finally say: 'Now go back through your organiser. Circle in red any effect you think cannot be reversed — like chapati that cannot go back to dough.' [Allow 2 minutes.]		
Explain Phase 📹 **VIDEO:** [Jamestown - the impact of tobacco](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemistry of Life (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 receive a CER (Claim–Evidence–Reasoning) frame card. The prompt is: 'ARGUMENT PROMPT: A form-three student in Kericho says, "I only smoke shisha made from natural fruit flavours — it cannot harm me because it is natural, like eating a mango." Use chemistry to argue whether this student is correct or incorrect.' Groups have 12 minutes to write a group CER response: one CLAIM (is the student correct?), at least THREE pieces of EVIDENCE from the video or organiser, and REASONING that explains how the chemistry of tar, nicotine, or carbon monoxide connects the evidence to the claim. One member acts as scribe, one as timekeeper, one as presenter, one as 'devil's advocate' (must challenge at least one piece of evidence before it is accepted by the group). Groups share with the class via a 90-second spoken presentation — no reading aloud, only speaking from notes. After each group, the class gives one 'warm' (strength) and one 'wonder' (question or challenge) on a sticky note.	Distribute CER cards and read the Kericho student prompt aloud. Say: 'This is a real argument you might hear from a friend or family member. Your job is not to be preachy — your job is to use chemistry. A chemist does not say "it is bad for you" — a chemist says "here is the evidence and here is the mechanism." You have 12 minutes.' [Circulate to each group. At the 6-minute mark, check that each group has a claim. Prompt stuck groups: 'What does carbon monoxide do to haemoglobin? Is that a reversible process? How does that connect to your claim?'] Before presentations, say: 'Presenters — no reading. Speak to us as if we are the Kericho student's parents and you are the chemistry teacher.' After each 90-second presentation, direct the class: 'Write one warm and one wonder on your sticky note — you have 60 seconds.' Cold-call 2 sticky-note authors to share their 'wonder' aloud. [WAIT TIME: 10 seconds after asking for wonders.] Close by saying: 'Look back at your prediction from Phase 1. Has the chemistry changed your mind? Update your notebook — write one sentence beginning: "The chemistry shows that..."'	Strategy: Claim–Evidence–Reasoning (CER) with Devil's Advocate Role. CER is an NGSS Science and Engineering Practice (Engaging in Argument from Evidence) that trains learners to distinguish observation from inference and to demand mechanistic explanation — not just correlation. The devil's advocate role internalises peer review and prevents groupthink. Anchoring the prompt in a familiar Kenyan social context (a peer in Kericho defending shisha) makes the argument personally relevant and prepares learners to apply this reasoning outside the classroom, fulfilling the Citizenship core competency.	Observable indicator: Each group's written CER contains (i) an explicit claim (correct/incorrect), (ii) at least one piece of evidence naming a specific chemical (tar, nicotine, or carbon monoxide) and its effect, and (iii) reasoning that uses the word 'because' or 'therefore' to connect evidence to claim. Teacher collects CER cards at the end of the lesson for written feedback. Formative red flag: Groups whose reasoning is only 'smoking is bad' without a named chemical mechanism — teacher writes on their card: 'Which chemical causes this? What does it do at the molecular level?'
Driving	The full class gathers facing the	Hand out green and orange sticky	Strategy: Two-Colour DQB Update	Observable indicator: Every

Question Board (DQB) Creation 📹 **VIDEO:** Jamestown - the impact of tobacco Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	DQB (a wall display or board maintained since Lesson 1). The teacher reads aloud the two driving question threads that this lesson addresses: (1) 'How does Chemistry explain what happens to the matter around us every day?' and (2) 'How does drug and substance use affect our health and well-being?' Learners individually write one new sticky note for each category: GREEN sticky = 'Something I can now explain using chemistry from today.' ORANGE sticky = 'A question this lesson raised that I still cannot answer.' Learners place their stickies on the DQB. The teacher reads 3–4 orange stickies aloud and the class votes (thumbs up) on which unanswered question they most want to pursue. The top-voted question is circled on the DQB and carried into Lesson 9.	notes. Say: 'Green note — what can you NOW explain using chemistry that you could NOT explain at the start of Lesson 1? It must be a chemistry explanation, not just a fact. Orange note — what new question did today's evidence raise?' [Allow 3 minutes of silent writing.] As learners place stickies, read 3–4 orange notes aloud in a neutral voice without identifying the authors. Ask: 'Thumbs up for the question you most want us to answer in our next lessons.' Count thumbs visibly. Circle the winning question. Say: 'This question goes on our DQB and travels with us. A good scientist is never satisfied — every answer opens a new question.' [WAIT TIME: 12 seconds after reading each orange note before asking for thumbs.]	(Explain + Wonder). The two-colour system separates consolidation (green) from curiosity (orange), preventing the DQB from becoming only a summary board. Reading orange stickies publicly validates intellectual uncertainty as a scientific virtue — directly modelling NGSS Practice 1 (Asking Questions). The class vote creates shared ownership of the inquiry direction and builds continuity across the 10-lesson sequence.	learner places at least one green and one orange sticky note. Teacher scans green notes for quality — a valid green note names a chemical mechanism (e.g., 'Carbon monoxide binds to haemoglobin and stops oxygen reaching cells — that is chemistry explaining a body change'). A weak green note says only 'smoking is bad' — teacher flags this learner for a brief check-in at the start of Lesson 9. Orange notes that reveal misconceptions (e.g., 'If nicotine is natural, why can't we just remove it?') are photographed by the teacher for use as a discussion prompt in Lesson 9.
Model Building Phase 📹 **VIDEO:** Jamestown - the impact of tobacco Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemistry of Life (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative unit model — a diagram they have been building since Lesson 1 that began with the chapati-dough phenomenon. The model currently shows: (i) raw dough → heat → chapati (irreversible physical/chemical change); (ii) matter defined; (iii) branches of chemistry; (iv) drugs, dosage, and prescription concepts from Lesson 7. Today, learners add a NEW PANEL to their model labelled: 'Chemistry explains irreversible changes in LIVING MATTER.' Inside this panel they draw a simple lung or blood vessel diagram and annotate it with the three chemicals (tar, nicotine, CO) and their specific irreversible effects. They then draw an arrow connecting this new panel back to	Distribute learners' cumulative model portfolios (or direct them to the relevant page in their notebooks). Say: 'Your model started with chapati. Today it grows. Add a new panel — I want to see the three chemicals, their effects, and a VISIBLE CONNECTION back to your chapati diagram. The connection should be a sentence — not just an arrow.' [Allow 8 minutes of individual model work. Circulate and prompt: 'Is the damage from tar reversible? How is that similar to or different from what happens to the dough?' and 'What branch of chemistry would study nicotine's effect on nerve cells?'] After 8 minutes: 'Find one person you have NOT worked with today. Share your model for 2 minutes —	Strategy: Cumulative Annotated Model Revision with Cross-Domain Linking. The cumulative model is the lesson sequence's primary sensemaking artefact — it externalises conceptual growth across all 10 lessons and forces learners to find structural parallels between domains (food chemistry and biochemistry of harm), which is the hallmark of deep disciplinary understanding. The cross-domain linking sentence requires learners to articulate the unifying principle of the unit — that chemistry governs all matter — moving them from isolated facts to integrated conceptual understanding. This directly fulfils NGSS Practice 2 (Developing and Using Models).	Observable indicator: The new model panel contains (i) at least two of the three chemicals named and their effects annotated, (ii) a visible link (arrow + sentence) to the chapati panel, and (iii) the linking sentence includes the word 'irreversible' or an equivalent. Teacher photographs 4–5 model pages at the end of class to use as exemplars and non-exemplars at the start of Lesson 9. A model with no connection to the chapati panel indicates the learner has not yet grasped the unit's anchoring principle and requires a targeted prompt at the start of Lesson 9.

	the chapati panel and write a linking sentence: 'Just as heat causes irreversible changes in chapati dough, tobacco chemicals cause irreversible changes in body tissue — because chemistry governs ALL matter, living and non-living.' Learners share their updated model with one new partner (not their usual group) for 2 minutes of peer feedback.	your partner must ask one question.' [Time the 2-minute partner share strictly.] Close with a whole-class share: cold-call 2 learners to read their linking sentence aloud. [WAIT TIME: 10 seconds after each.] End by saying: 'Next lesson we go deeper into substance use and your rights. Your model will grow again.'		

D. TEACHER REFLECTION

After the lesson, the teacher should consider the following questions: (1) Did learners' CER arguments name specific chemical mechanisms (tar, nicotine, CO), or did they rely on general statements like 'smoking is bad'? If the latter, the Explain phase needs more structured scaffolding in the next lesson — consider pre-teaching the three chemicals using a quick matching card activity before the next evidence-gathering task. (2) Were learners able to draw the conceptual link between irreversible chemical changes in chapati dough and irreversible damage in body tissue? If this connection was absent from most models, begin Lesson 9 with a 5-minute whole-class model comparison using one strong and one weak example (anonymised) projected on the board. (3) What unanswered questions appeared most frequently on orange DQB stickies? Use the top-voted question as the opening provocation for Lesson 9. (4) Were there learners who appeared personally affected by the tobacco discussion (e.g., a parent who smokes)? Follow up individually and sensitively before the next lesson — affirm that chemistry helps us understand and protect, not judge. (5) Did the devil's advocate role in the CER groups actually generate productive challenge, or did groups skip it? If skipped, assign the role more explicitly in Lesson 9 by giving the devil's advocate a different-coloured pen and a specific instruction card.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners watched a video showing how tobacco smoke releases tar, nicotine, and carbon monoxide into the body, and annotated a body-system diagram with the specific effects of each chemical on lungs, blood vessels, and the nervous system.
What did I learn?	Learners learned that 'natural' does not mean 'safe' — tobacco is a natural plant, yet its chemicals cause specific, largely irreversible damage at the molecular level; tar coats and destroys lung tissue, nicotine creates chemical dependency by acting on nerve receptors, and carbon monoxide binds to haemoglobin and blocks oxygen transport, just as heat irreversibly transforms chapati dough into a fundamentally different substance.
How does this explain the phenomenon?	Learners explained, using a Claim–Evidence–Reasoning frame, why the statement 'shisha is natural so it is safe' is chemically incorrect, connecting evidence from the video to the anchoring phenomenon: Chemistry governs all matter — living and non-living — and the same principle that explains why chapati dough cannot return to its original form explains why tobacco damage to lung tissue and haemoglobin cannot be easily undone.

LESSON 9 (80 minutes): Consumer Protection and the Safe Learning Environment

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To connect Chemistry knowledge to consumer protection and personal safety by examining real Kenyan product labels and brainstorming learners' rights and responsibilities in a safe, healthy learning environment.
Knowledge	Learners will identify and interpret KEBS quality marks, expiry dates, nutritional information, preservatives, safety warnings, and cautionary messages on labels of manufactured Kenyan products (food, cleaning agent, medicine).
Skills	Learners will analyse product labels for consumer protection information, evaluate chemical safety risks in everyday and laboratory contexts, and construct arguments linking Chemistry knowledge to responsible personal decision-making.
Attitudes	Learners will advocate for a safe and healthy learning environment and demonstrate responsible citizenship by recognising their rights and responsibilities as consumers and as science learners.
Key Inquiry Question	How does drug and substance use affect our health and well-being?
Purpose in Storyline	In Lessons 1–8 learners built a foundation: Chemistry defines and explains changes in matter, its branches and careers connect to real industries, physical and chemical changes govern everyday transformations (including our chapati phenomenon), and drugs interact with the body at a molecular level. Lesson 9 now turns that knowledge outward — learners become critical readers of the chemical world around them. The chapati on the jiko used wheat flour bought from a shop; that packet carried a KEBS mark, a nutritional label, and an expiry date. Today learners ask: what chemical information must be communicated on products, and what rights and responsibilities does that create? This bridges consumer protection (a Chemistry application) to lab safety and the broader driving question about making safer, smarter choices.
Safety Notes	No hazardous chemicals are used in this lesson. All product samples (food packets, cleaning agent containers, medicine packaging) must be empty, sealed, or handled in their original, closed packaging. Learners must not open, taste, or inhale contents of any sample. If a cleaning agent container is used, ensure it is rinsed and dried before the lesson. Remind learners that knowledge of chemical labels is itself a safety skill — misreading or ignoring a label can cause harm.

B. LESSON OVERVIEW

Learners work in small groups to examine labels on 3–4 real Kenyan manufactured products — a packet of Jogoo maize flour, a bottle of Jik bleach (or similar household cleaner), a strip/box of paracetamol (or an ORS sachet), and optionally a tin of Kimbo or packet of Ketepa tea. Each group reads and records consumer protection information: KEBS quality marks, expiry/manufacture dates, ingredient lists, preservative codes (E-numbers), nutritional panels, dosage instructions, and safety/caution messages. Groups then participate in a structured Socratic discussion to brainstorm their rights (to accurate information, to quality products, to safety warnings) and responsibilities (reading labels, correct dosage, proper storage, not misusing lab or household chemicals). The lesson culminates in learners connecting all accumulated evidence to the driving question, updating the Driving Question Board, and adding a final layer to their cumulative matter-change model that now includes the human body as a chemical system that must be protected.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Reactants, Products and Leftovers Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **PDF:** Introducing Green Ch..:Teacher Background Information: The 12 Principles of Green Chemistry Source: MIT Blossoms 🔍 Search ARES for similar readings	The teacher holds up a closed packet of Jogoo maize flour (or displays a projected image of it) and asks: 'Before we read a single word on this packet — what chemical information do you predict is printed here, and why would a manufacturer bother to include it?' Learners write individual predictions in their notebooks under two columns: 'Information I predict is there' and 'Reason it matters chemically.' They then share predictions with a shoulder partner for 2 minutes. The teacher selects 4–5 learners via cold-call to share, recording key predicted categories on the board (ingredients, expiry, nutritional info, safety warnings, quality mark). The teacher connects back to the phenomenon: 'When someone bought the maize flour to make chapati dough, they trusted that packet. What were they trusting Chemistry to guarantee?'	1. Display the Jogoo flour packet prominently. Say: 'Do not read the label yet — just look at the packet and predict.' Allow 3 minutes of silent individual writing. 2. Say: 'Turn to your shoulder partner. Compare your two prediction lists. You have 2 minutes.' [WAIT — allow full 2 minutes of paired talk.] 3. Cold-call exactly 5 learners using the class register: 'Amina, what is one thing you predicted? Thank you. Kamau, what did you add that Amina did not mention?' Record each unique category on the board as a T-chart: 'Predicted Information | Why It Matters Chemically.' 4. After cold-calls, ask the whole class: 'Raise your hand if you predicted a KEBS quality mark.' Pause 10 seconds. 'Who predicted an E-number or preservative code?' Pause 10 seconds. Note gaps — these become the inquiry targets. 5. Pose the phenomenon bridge question: 'Whoever bought flour to make today's chapati trusted that packet. What is Chemistry promising you through that label?' Allow 15 seconds of wait time before taking 2–3 responses.	Prediction-to-Evidence Anchoring: By committing predictions to paper before examining labels, learners activate prior knowledge from Lessons 1–8 (matter composition, chemical changes, drug safety) and create a cognitive 'expectation gap' that drives motivated observation in Phase 2. Recording predictions publicly on the board also makes thinking visible and gives the teacher real-time diagnostic data about misconceptions.	Observable indicator: Every learner has a written two-column prediction list in their notebook before partner talk begins. The teacher circulates during individual writing and looks for at least 3 predicted categories per learner. During cold-calls, listen for whether learners connect label information to Chemistry concepts (e.g., 'expiry date tells us the chemical reactions inside have not degraded the product yet') — this reveals depth of understanding from prior lessons. Flag learners who list only surface features (e.g., 'price', 'brand name') for targeted questioning in Phase 2.
Observe Phase 📹 **VIDEO:** Video: Reactants, Products and Leftovers Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **PDF:**	The teacher distributes one product sample per group of 4–5 learners, rotating samples after 8 minutes so every group examines at least two different product types. The four sample types are: (A) Jogoo maize flour packet, (B) Jik bleach bottle (empty/rinsed and dry), (C) Paracetamol box/strip with package insert, (D) Ketepa tea packet or Kimbo tin. Each group uses a structured Label Analysis Recording Sheet with the	1. Before distributing samples, say clearly: 'These containers are to be read — not opened, not tasted, not sniffed. Our safety rule today: eyes and hands only on the outside.' Pause and ensure all learners signal understanding. 2. Distribute one product per group. Say: 'You have 8 minutes with this product. Fill in every row of your recording sheet. Go.' Set a visible timer. Circulate briskly — visit every group at least once. Prompt with:	Structured Observation with a Recording Framework (Science and Engineering Practice: Analysing and Interpreting Data): The Label Analysis Recording Sheet transforms a familiar consumer object into a data-collection instrument. By imposing scientific categories (chemical names, safety codes, reaction warnings) onto everyday packaging, learners experience how Chemistry is embedded in	Observable indicator: Every learner's recording sheet has at least 8 of 10 rows completed after both rotations. The teacher checks sheets during circulation — specifically look for: (a) correct identification of the KEBS mark on at least one product, (b) at least one E-number or chemical name written with its function (not just its code), (c) a genuine question written in row (x) that is chemical in nature (not logistical). During the

Introducing Green Ch..:Teacher Background Information: The 12 Principles of Green Chemistry Source: MIT Blossoms 🔍 Search ARES for similar readings	following columns: (i) Product name and manufacturer, (ii) KEBS mark — present? Describe it, (iii) Manufacture date / Expiry date, (iv) Ingredients / Active chemicals listed, (v) Preservatives or additive codes (E-numbers), (vi) Nutritional information (if present), (vii) Dosage / Usage instructions, (viii) Safety warnings / Caution messages, (ix) Disposal instructions (if present), (x) One question this label raises for us. Learners record observations carefully in their sheets. The teacher circulates, prompting deeper reading without giving answers. After both rotations, the class reconvenes. Each group's recorder shares their most surprising finding — the teacher scribes key chemical terms on the board (e.g., Sodium hypochlorite, Paracetamol 500mg, Ascorbic acid/E300, 'Keep out of reach of children', 'Do not exceed stated dose').	'What does that E-number stand for — is it a preservative, a colour, an antioxidant?' and 'Where exactly is the KEBS mark — what does it look like?' 3. After 8 minutes, say: 'Rotate — pass your product to the next group clockwise. You have another 8 minutes.' Repeat circulation. 4. Reconvene. Ask each group's recorder: 'Give me the one chemical name on your label that surprised your group most.' Cold-call all 5–6 groups. Write each chemical name on the board. Ask: 'What branch of Chemistry studies this substance?' (Linking to Lesson 2 — branches of Chemistry.) 5. Point to the bleach chemical name (Sodium hypochlorite) and ask: 'This is a chemical change agent — if we mixed this with vinegar at home, what might happen? Why does the label warn against it?' Allow 15 seconds wait time. Take 3 responses. Record the key insight: 'Labels communicate chemical reaction risks — reading them is a Chemistry skill.'	manufactured products. Rotating samples ensures all learners encounter both food-chemistry contexts (flour, tea) and hazard-chemistry contexts (bleach, medicine), building a complete picture of consumer protection.	share-out, listen for whether learners spontaneously connect chemical names to branches of Chemistry from Lesson 2 — this is the key cross-lesson integration indicator.
Explain Phase 📹 **VIDEO:** Video: Reactants, Products and Leftovers Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **PDF:** Introducing Green Ch..:Teacher Background Information: The 12 Principles of	The teacher leads a whole-class Socratic seminar structured around two interconnected questions: (1) 'As a consumer, what rights does Chemistry give you — and what responsibilities does it place on you?' and (2) 'As a Chemistry learner in a school lab, what rights and responsibilities do you have to keep your learning environment safe?' Learners first brainstorm individually (2 minutes), then share in groups of 4–5 (4 minutes), building a group Rights-and-Responsibilities T-chart. Groups then present their T-charts to the class. The teacher synthesises responses into a class	1. Write the two Socratic questions on the board. Say: 'In your notebook, write at least three rights and three responsibilities — one column each. You have 2 minutes. Silent, individual thinking.' [WAIT — enforce silence for full 2 minutes.] 2. Say: 'Now, in your group of four, build a combined T-chart on one sheet of paper. You have 4 minutes. Every person must contribute at least one item.' Circulate; if a group is listing only generic items ('be safe'), prompt: 'Can you make that more specific to Chemistry? What specific chemical knowledge gives you that responsibility?' 3. Cold-call one	Socratic Seminar with Rights-and-Responsibilities Mapping (Science and Engineering Practice: Constructing Explanations; PCI — Life Skills: Safety and Security; PCI — Consumer Protection): This strategy requires learners to use Chemistry knowledge as the justification for civic rights and personal responsibilities — not just to list rules, but to explain them chemically. Mapping rights to specific Chemistry concepts (additives, reactions, dosage, hazardous products) makes abstract consumer protection law concrete and science-based, deepening the competency of	Observable indicator: Each group's T-chart contains at least 4 rights and 4 responsibilities, with at least 3 of these explicitly linked to a Chemistry concept (not just a general safety rule). During the Socratic seminar, the teacher listens for learners using Chemistry vocabulary from previous lessons (e.g., 'chemical reaction', 'preservative', 'dosage', 'toxic', 'KEBS standard', 'molecular', 'substance'). A learner who can explain WHY a safety warning exists in chemical terms (not just that it exists) demonstrates the target level of understanding. Collect 2–3 group

Green Chemistry Source: MIT Blossoms 🔍 Search ARES for similar readings	master T-chart on the board, explicitly linking each right/responsibility to a Chemistry concept: e.g., 'The right to know what preservatives are in your food' links to knowledge of chemical additives and their reactions in the body; 'The responsibility not to mix household chemicals' links to knowledge of chemical reactions and hazardous products; 'The right to accurate dosage information' links to drug pharmacology from Lesson 8; 'The responsibility to read lab safety instructions' links to the nature of chemical hazards. Learners copy the master T-chart into their notebooks. The teacher then returns to the chapati phenomenon: 'The flour that became chapati — was it protected by Chemistry before it even reached the jiko? How?'	spokesperson per group to share their most important right and most important responsibility. After each group, ask the class: 'Does anyone want to challenge or add to what [Name]'s group said?' Allow 10 seconds wait time each time before prompting further. 4. Build the master T-chart on the board in real time. After every 3–4 entries, ask: 'Which branch of Chemistry from Lesson 2 is most relevant to this right or responsibility?' Cold-call 3 different learners each time. 5. Ask the phenomenon bridge question: 'The flour became chapati — but before it reached the jiko, Chemistry was already protecting you. How?' Allow 15 seconds wait time. Expect responses about KEBS quality marks, preservatives, expiry dates, ingredient safety. Affirm and extend: 'Chemistry does not just happen on the jiko — it is present all the way from the farm, through the factory, through the label, to your body.'	Citizenship.	T-charts at end of lesson for written feedback.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Reactants, Products and Leftovers Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **PDF:** Introducing Green Ch..:Teacher Background	The teacher returns to the class Driving Question Board displayed at the front of the room. Learners review the two driving questions: (1) 'How does Chemistry explain what happens to the matter around us every day?' and (2) 'Why does understanding Chemistry help us make safer, smarter choices in our lives?' The teacher reads aloud each sticky note or card that learners have added across Lessons 1–8, briefly narrating the evidence journey: definitions of Chemistry and matter → branches → careers → physical vs chemical changes → chapati as a chemical change → states of matter → properties of matter → drugs and	1. Stand at the DQB. Say: 'Let us walk through the evidence we have gathered across 9 lessons.' Narrate each phase in 30 seconds, pointing to relevant cards/notes already on the board. This takes approximately 3 minutes total. 2. Distribute two sticky notes (different colours: one for 'Evidence', one for 'New Question') to every learner. Say: 'On your Evidence note, write one thing from today — one thing you observed on a label or in our discussion — that directly answers part of our driving question. On your Question note, write one new question today's lesson raised for you.' Allow 3 minutes of individual	Driving Question Board Curation (Science and Engineering Practice: Obtaining, Evaluating, and Communicating Information): The DQB is not merely a display — it is a cumulative argument board. By physically adding evidence and questions each lesson, learners experience how scientific understanding is built incrementally. In Lesson 9, the act of narrating the full 8-lesson evidence journey before adding new notes gives learners a metacognitive map of their own learning, strengthening the coherence of the unit's storyline and preparing them for the Lesson 10 synthesis.	Observable indicator: Every learner posts exactly two sticky notes — one 'Evidence' and one 'New Question.' The teacher reads through all posted evidence notes after class and categorises them: (a) notes that make a direct, Chemistry-grounded connection to the driving question (target level), (b) notes that make a general connection without Chemistry vocabulary (developing level), (c) notes that are off-topic or too vague (needs support). This categorisation informs differentiated support during Lesson 10's synthesis activity.

Information: The 12 Principles of Green Chemistry Source: MIT Blossoms 🔍 Search ARES for similar readings	the body. Now learners write new DQB contributions on sticky notes focused on today's evidence: What new evidence from product labels answers part of the driving question? What new questions has this lesson raised that still need answering? Learners post their notes, grouping them under two headers: 'Evidence that answers our driving question' and 'New questions we still have.' The teacher reads 5–6 notes aloud and facilitates a 3-minute whole-class discussion: 'How does reading a KEBS label on a packet of flour connect to why Chemistry matters in our daily lives?' The teacher highlights that Lesson 10 will synthesise all evidence into a final poster and presentation, completing the answer to the driving question.	writing. 3. Invite all learners to post their notes on the DQB under the correct header. Allow 2 minutes for posting. 4. Read 5–6 notes aloud (choose a mix of strong evidence statements and genuinely interesting new questions). For each, ask: 'Who agrees this is strong evidence? Thumbs up if yes.' Then: 'Who can explain in one sentence WHY this is evidence for our driving question?' Cold-call 2 learners per selected note. 5. Close by saying: 'In Lesson 10, you will use ALL of this evidence — 9 lessons of it — to build a final poster that answers our driving question completely. Today was the last piece of evidence. Everything is now on the board.'		
Model Building Phase 📹 **VIDEO:** Video: Reactants, Products and Leftovers Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **PDF:** Introducing Green Ch..:Teacher Background Information: The 12 Principles of Green Chemistry Source: MIT Blossoms 🔍 Search ARES for similar	Learners open their cumulative model notebooks — the same model they have been revising since Lesson 1 (which began as a simple diagram of the chapati phenomenon). In this lesson, learners add a final outer layer to their model: the 'Chemistry as Protection' layer. They draw or annotate to show: (1) The chapati's journey — flour packet (with KEBS mark and preservatives) → dough (physical mixing, chemical potential) → chapati (irreversible chemical change on the jiko) — and add labels showing where consumer protection Chemistry was operating at each stage. (2) An arrow pointing from 'Chemistry knowledge' to 'Safer choices' with at least three specific examples from today's lesson (e.g., reading expiry dates, understanding	1. Say: 'Open your cumulative model. Look at what you drew in Lesson 1. In eight lessons, how much has your model grown? Take 30 seconds to flip through your model notebook silently.' Allow 30 seconds. 2. Say: 'Today we add the final outer layer — Chemistry as Protection. You have 10 minutes. Your model must show: the chapati's journey with consumer protection Chemistry labelled at each stage; an arrow from Chemistry knowledge to safer choices with three examples; and the human body as a chemical system protected by label-reading. Go.' Set a timer. Circulate and narrate what you see: 'I notice Wanjiku has added the KEBS mark to her flour packet — excellent. Can you add the specific chemical that acts as a preservative in that flour?' 3. After	Cumulative Model Revision — 'Chemistry as Protection' Layer (Science and Engineering Practice: Developing and Using Models): Adding a final outer 'protection' layer to a model that began with a single kitchen observation makes the unit's conceptual arc visible and concrete. The model now encodes the entire argument: Chemistry explains matter changes (chapati), Chemistry is applied in industries (food, pharmaceutical), Chemistry knowledge enables safer choices (consumer protection, drug safety). The two-sentence Model Narrative forces learners to articulate the connection between their visual model and the driving question, building scientific communication skills.	Observable indicator: The updated model contains all three required elements: (a) the chapati journey with consumer-protection labels, (b) the 'Chemistry knowledge → safer choices' arrow with at least three specific examples, (c) the human body inset from Lesson 8. The two-sentence Model Narrative is present and uses at least two Chemistry vocabulary terms. During peer feedback, the teacher listens for whether feedback is Chemistry-specific (target level) or only aesthetic ('your drawing is neat'). Collect 4–5 model notebooks at random after class to assess depth of integration before Lesson 10.

readings	preservative codes, following medicine dosage instructions, not mixing bleach with other cleaners). (3) A small inset diagram showing the human body as a chemical system (from Lesson 8) and how correct dosage/label-reading protects that system. Learners write a two-sentence 'Model Narrative' below their diagram: 'My model now shows… / This connects to the driving question because…' The teacher selects 3 volunteers to share their updated models under the visualiser or by holding them up, and facilitates peer feedback using the stem: 'I can see [Chemistry concept] in your model — I would suggest adding [missing element].'	10 minutes, ask for 3 volunteers. Under the visualiser, each volunteer shows their model for 90 seconds. The class gives feedback using the stem: 'I can see [concept] — I suggest adding [element].' Cold-call 2 peers per presenter for feedback. 4. Ask all learners to write their two-sentence Model Narrative. Allow 2 minutes. 5. Close the lesson by saying: 'In Lesson 1, your model showed one thing: chapati dough changing on a jiko. Today, your model shows Chemistry protecting matter — from the farm, to the factory, to the label, to the jiko, to your body. That is what Chemistry does every day. That is why it matters.'		

D. TEACHER REFLECTION

After Lesson 9, reflect on the following questions before planning Lesson 10:

1. LABEL LITERACY: Could learners independently identify and explain chemical information on product labels — including E-numbers, active ingredients, and hazard warnings — using Chemistry vocabulary? Or did they describe labels only in consumer/marketing terms? If the latter, build a brief vocabulary recap into the opening of Lesson 10.

2. RIGHTS AND RESPONSIBILITIES DEPTH: Did learners' T-charts contain Chemistry-grounded justifications for rights and responsibilities, or were entries generic safety slogans? If groups struggled to link rights to Chemistry concepts, consider providing a sentence starter scaffold in Lesson 10: 'I have the right to… because Chemistry tells us that…'

3. DQB EVIDENCE QUALITY: Review the sticky notes posted on the DQB. Are learners making direct, specific connections between today's evidence (labels, consumer protection) and the driving question? Or are connections vague? Use the strongest 3 evidence notes as exemplars to open Lesson 10's synthesis activity.

4. MODEL INTEGRATION: Did learners successfully connect the 'Chemistry as Protection' layer to their chapati phenomenon origin story? The most common gap at this stage is learners treating the model's layers as separate topics rather than one integrated argument. In Lesson 10, ask learners to draw a single 'red thread' line through all layers of their model showing how one idea — Chemistry explains and protects matter — runs through every lesson.

5. SAFETY CULTURE: Did learners handle product samples responsibly and follow the 'eyes and hands only' rule without prompting? This is itself a formative indicator of the sub-strand's attitudinal outcome: advocating for a safe learning environment. Note any learners who needed repeated reminders for follow-up mentoring.

6. LESSON 10 READINESS: Lesson 10 is the final synthesis lesson — learners will develop a poster answering the driving question using all 9 lessons of evidence. Before that lesson, ensure: (a) the DQB is fully updated and visible, (b) learners have their complete cumulative model notebooks, (c) the teacher has reviewed 4–5 model notebooks and prepared specific written feedback to return at the start of Lesson 10.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined labels on real Kenyan manufactured products — including Jogoo maize flour, Jik bleach, and paracetamol packaging — recording KEBS quality marks, expiry dates, ingredient lists, E-number preservative codes, nutritional panels, dosage instructions, and safety/caution warnings on a structured Label Analysis Recording Sheet.
What did I learn?	Learners learned that Chemistry is embedded in every manufactured product through quality standards, preservative chemistry, dosage pharmacology, and hazard communication; that consumer protection rights (to accurate information, quality assurance, safety warnings) are grounded in Chemistry knowledge; and that reading and interpreting chemical labels is itself a Chemistry skill that enables safer, smarter daily choices.
How does this explain the phenomenon?	Learners explained how Chemistry knowledge creates both rights (e.g., the right to know what preservatives are in food, the right to accurate dosage information) and responsibilities (e.g., the responsibility to read safety warnings, to avoid misusing lab or household chemicals, to follow correct dosage), and connected the chapati phenomenon's flour-to-product journey to the full chain of consumer protection Chemistry — from farm to factory to label to jiko to body.

LESSON 10 (80 minutes): Project: Poster on Drug & Substance Use — Unit Synthesis & Final Explanation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate the entire sub-strand by completing the community awareness poster project on drug and substance use, finalising the concept-map model, completing the Driving Question Board, and constructing a full evidence-based answer to the unit's driving question.
Knowledge	Learners demonstrate understanding of: the definition of Chemistry as the study of matter and its changes; the branches of Chemistry and their real-world roles; the importance of Chemistry in agriculture, pharmaceutical, food, and other industries; the meaning of drug, prescription, dosage, and substance use; and the effects of drug and substance use on daily life and health.
Skills	Learners apply communication and collaboration skills to finalise and present posters; synthesise evidence from across the unit into a coherent final explanation; critically evaluate their Lesson 1 predictions against final understanding; and revise their concept-map model to reflect complete unit learning.
Attitudes	Learners advocate for a safe and healthy learning environment; demonstrate respect for peers' contributions; show self-efficacy in presenting findings; and display responsible citizenship by communicating evidence-based warnings about drug and substance misuse.
Key Inquiry Question	1. What is Chemistry and why is it important in our daily lives? 2. How does drug and substance use affect our health and well-being?
Purpose in Storyline	This is the capstone lesson of the unit. Learners return to the chapati phenomenon one final time — not as a puzzle, but as a fully explained anchor. They connect every thread of the sub-strand (definition of Chemistry, branches, careers, matter, drug and substance use) back to that single image of dough transforming on a jiko. The poster project gives learners a tangible, community-facing product that embodies Chemistry's power to inform safer choices. The DQB is closed, the concept map is displayed, and learners leave understanding that Chemistry is not an abstract school subject — it is the science that explains their world, from the kitchen jiko to the pharmacy shelf.
Safety Notes	No hazardous materials are used in this lesson. Where poster content references drug substances, remind learners to use respectful, evidence-based language and to avoid stigmatising individuals. Ensure that any personal disclosures made during reflective discussion are handled with sensitivity and referred to the school counsellor if necessary.

B. LESSON OVERVIEW

In this final lesson of Sub-Strand 1.1, learners work in groups to complete and present their community awareness posters on the risks of drug and substance use — the KICD-mandated project. Poster content draws directly on evidence gathered across all ten lessons. After gallery-walk presentations, the class gathers for a whole-group Final Explanation session: they revisit their very first Lesson 1 predictions about the chapati phenomenon, compare them with their current understanding, and co-construct a full written answer to the driving question. The concept-map model (built incrementally from Lesson 1) is finalised and displayed. The Driving Question Board is reviewed: answered questions are formally crossed off, and remaining questions are flagged for Sub-Strand 1.2 and beyond. The lesson closes with individual written reflections and a teacher-led celebration of learning growth.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	The teacher projects (or re-draws	Display or sketch the chapati-on-	Metacognitive Prediction Revisit —	Observable indicator: Each learner

📹 **VIDEO:** [Coordinate plane word problem examples] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [chemistry-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	on the board) the exact Lesson 1 anchor image: a ball of raw chapati dough being placed on a hot jiko. Learners individually retrieve their Lesson 1 prediction cards/notebooks and re-read what they wrote at the very start of the unit. They then silently write two sentences: (1) 'In Lesson 1, I thought… because…' and (2) 'Now I think… because…'. Pairs share briefly (1 minute) before the class reconvenes. This metacognitive step activates the full arc of their learning and sets up the Final Explanation.	jiko image prominently. Say verbatim: 'Before we do anything else today, I want you to travel back in time to Lesson 1. Take out your prediction from that first day — the one you wrote before we had studied anything.' Allow 2 minutes silent retrieval and re-reading. Then say: 'Now write two sentences — what did you believe then, and what do you understand now? Be specific — use the vocabulary you have learned.' Allow 4 minutes silent writing. Cold-call 4 learners (mix of confidence levels) to share — ask each: 'What is the biggest single thing Chemistry has helped you understand about the chapati?' Apply 10-second WAIT TIME before each cold-call response. Record key before/after contrasts on the board in two columns labelled 'Lesson 1 Thinking' and 'Lesson 10 Thinking'.	By explicitly comparing initial naïve ideas with current evidence-based understanding, learners make their own intellectual growth visible. This strategy consolidates long-term retention and gives the Final Explanation its emotional and cognitive anchor.	produces two written sentences that correctly contrast a Lesson 1 misconception or incomplete idea with a Lesson 10 Chemistry-vocabulary-rich explanation. Teacher scans notebooks during cold-call discussion. Red flag: learners who cannot identify a change in thinking — pull them into a brief pair with a peer before Phase 3.
Observe Phase 📹 **VIDEO:** [Coordinate plane word problem examples] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [chemistry-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Groups (formed in Lesson 9) post their completed community awareness posters around the classroom walls — these address the risks of drug and substance use, using Chemistry vocabulary (molecular interaction, dosage, prescription, substance use, matter, chemical change). The class conducts a structured Gallery Walk (15 minutes): each group leaves one 'Ambassador' at their poster while other members rotate. Rotating learners use a sticky-note protocol — green sticky = one piece of evidence used correctly, yellow sticky = one question or suggestion. After the walk, Ambassadors read back the sticky notes to their group. Groups have 3 minutes to note any final edits. Teacher simultaneously	Before the Gallery Walk, say: 'As you visit each poster, you are acting as peer scientists reviewing evidence. Ask yourself: Is the claim supported? Is the Chemistry accurate? Is the advice practical for a family in Kericho or Kisumu or Nairobi?' Circulate during the walk — pause at 2–3 posters and ask the Ambassador a probing question, e.g. 'Your poster mentions chemical dependence — which branch of Chemistry helps explain how substances interact with body cells?' Apply 12-second WAIT TIME. After the walk, cold-call 3 Ambassadors to share the most interesting sticky-note feedback received. Say: 'These posters are not school work — they are public health tools. Chemistry gives them their	Peer Evidence Review (Gallery Walk with Sticky-Note Protocol) — Learners act as both communicators and critical consumers of evidence. Seeing multiple posters reinforces the breadth of Chemistry's relevance (pharmaceutical, food, health) and models the kind of evidence-based public communication that characterises scientific citizenship.	Observable indicator: Each poster contains a minimum of three Chemistry-specific terms used accurately, at least one Kenyan context (e.g., reference to NACADA warnings, local substance misuse patterns, pharmacy labelling), and a clear evidence-based caution statement. Teacher photographs posters for portfolio records. Green/yellow sticky-note distribution gives real-time peer assessment data.

	conducts an Evidence Audit: at each poster, they check that learners have cited at least one Chemistry branch, one real-world Kenyan context, and one specific health effect of drug/substance misuse.	credibility.' Time-keep strictly: Gallery Walk 15 min, Feedback 3 min, Edit window 3 min.		
Explain Phase 📹 **VIDEO:** [Coordinate plane word problem examples] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [chemistry-question-paper-1-kcse-2012] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	The class gathers in a full-group circle or U-shape. The driving question is displayed prominently: 'How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?' The teacher facilitates a structured co-construction of the Final Explanation, moving through three evidence threads: (1) The Chapati Thread — what Chemistry tells us about the transformation of dough on the jiko (matter, physical vs. chemical change, irreversibility); (2) The World Around Us Thread — how the branches of Chemistry explain phenomena in agriculture (Rift Valley farms), medicine (Nairobi hospitals), food (githeri, sukuma wiki, mandazi), manufacturing, and sport; (3) The Safer Smarter Choices Thread — how Chemistry knowledge about drugs, dosage, prescription, and substance effects empowers learners to make evidence-based decisions. Learners contribute orally; the teacher scribes a class Final Explanation paragraph on the board. Each learner then writes their own individual Final Explanation (minimum 8 sentences) in their notebooks.	Open with: 'Ten lessons ago, a chapati was cooking on a jiko and none of us could fully explain it. Today we can. Let us build that explanation together — one piece of evidence at a time.' Use a three-column scaffold on the board: 'What we observed | What Chemistry tells us | Why this matters for our lives.' Cold-call 6 different learners across the three threads (2 per thread) — use WAIT TIME of 15 seconds per response. After each contribution, press for precision: 'Can you name which branch of Chemistry explains that?' and 'Where in Kenya would a professional use that knowledge?' Scribe the co-constructed paragraph collaboratively, editing in real time with class input. Once the class paragraph is complete, say: 'Now write your own. This is your final evidence that you understand Chemistry — not just as a school subject, but as a tool for life.' Allow 10 minutes for individual writing. Circulate and prompt any learner who is stuck: 'Start with the chapati — what did Chemistry help you see that you could not see before?'	Three-Thread Final Explanation — Organising the explanation into three evidence threads (phenomenon, world, choices) prevents a superficial list and instead builds a coherent narrative. This mirrors how scientists construct explanations: grounded in observable evidence, interpreted through disciplinary concepts, and connected to human relevance. The individual writing step ensures every learner personally owns the explanation rather than free-riding on peers.	Observable indicator: Individual Final Explanation paragraphs correctly use a minimum of 6 of the following unit vocabulary terms: matter, chemical change, physical change, branch of Chemistry, pharmaceutical, dosage, prescription, substance use, irreversible, properties of matter. Teacher collects notebooks at the end of the lesson for summative review. Red flag: incomplete or vocabulary-absent paragraphs — these learners are identified for follow-up at the start of Sub-Strand 1.2.
Driving Question Board (DQB)	The Driving Question Board (built from Lesson 1 and added to across the unit) is brought to the	Stand at the DQB and say: 'This board has been our scientific compass for ten lessons. Every	DQB Closure Protocol — Publicly marking the status of each question externalises the class's	Observable indicator: The class correctly categorises at least 80% of DQB questions without teacher

Creation 📹 **VIDEO:** Coordinate plane word problem examples Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** chemistry-question-paper-1-kcse-2012 Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	front of the class. Working as a whole group, learners systematically review each question card. For each question, the class votes: 'Have we answered this fully? Partially? Not yet?' Answered questions are crossed off with a green marker — the learner who first raised the question has the honour of crossing it off. Partially answered questions receive an orange dot. Unanswered questions receive a blue arrow labelled 'Coming in Sub-Strand 1.2 and beyond.' Learners count: How many questions did we answer? How many new questions did Chemistry open up? The teacher introduces the idea that good science always generates more questions than it closes — this is celebrated, not lamented.	question on it was asked by you — let us see how far we have come.' Read each question card aloud. After each, pause 10 seconds and ask for a show of hands: 'Fully answered? Raise your hand.' Do not rush — honour each question. For questions being crossed off, invite the original asker: 'This was your question, [learner name]. Would you like to cross it off and tell us how Chemistry answered it?' For blue-arrow questions, say: 'This question is not unanswered because we failed — it is unanswered because it belongs to a deeper layer of Chemistry we are about to enter.' Point forward to Sub-Strand 1.2 (The Atom). Close with: 'Count the green crosses. Every one of those is a question your brain now has an answer to. That is Chemistry at work.'	collective intellectual journey. The distinction between 'answered,' 'partially answered,' and 'forwarded to future study' models authentic scientific practice — no field is ever fully closed. The ritual of the original asker crossing off their own question creates personal investment and celebrates curiosity as a scientific virtue.	correction. Learners can verbally justify why a question is 'answered' by citing a specific lesson's evidence. Teacher notes which blue-arrow questions cluster around atomic structure — these directly preview Sub-Strand 1.2 and inform the teacher's introduction to that sub-strand.
Model Building Phase 📹 **VIDEO:** Coordinate plane word problem examples Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** chemistry-question-paper-1-kcse-2012 Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Groups retrieve the concept-map model they have been building and revising since Lesson 1. They have a final 8-minute window to make last additions or corrections — particularly connecting the 'Drug and Substance Use' node to Chemistry branches (biochemistry, pharmaceutical chemistry) and to the 'Matter and its Changes' central node. Each group then selects one 'lead presenter' who gives a 90-second spoken tour of their final concept map to the class, explicitly stating: 'Our model changed from Lesson 1 to Lesson 10 in these ways…' Groups then display their finalised concept maps on the classroom wall alongside their posters. The teacher closes with a whole-class moment: learners stand, look at the displayed work, and the	Distribute concept maps and say: 'You have 8 minutes — final edits only. Ask yourself: Does our map show the connection between Chemistry, matter, and safer choices? Does the chapati phenomenon appear somewhere on this map? If not, add it now.' Circulate — prompt with: 'Where on your map does the pharmaceutical industry connect to drug safety?' and 'Which branch of Chemistry explains why chapati cannot become dough again?' After presentations, cold-call 2 groups to compare: 'Group A, Group B — what is the most important difference between your Lesson 1 model and your Lesson 10 model?' Apply 12-second WAIT TIME. For the closing gallery moment, say slowly and with deliberate pacing: 'Everything on	Cumulative Model Comparison (Lesson 1 vs Lesson 10) — Placing the Lesson 1 and Lesson 10 models side by side makes conceptual growth tangible and visible. The 90-second spoken tour requires learners to narrate change — a high-order metacognitive act that consolidates understanding far more deeply than passive review. Displaying models and posters together creates a public record of the class's scientific community work.	Observable indicator: Final concept maps include a minimum of 8 nodes correctly connected, with at least one explicit link between 'Chemistry branches,' 'matter and its changes,' 'drug and substance use,' and the chapati phenomenon. The 90-second tour is assessed using a simple 3-point rubric (1 = general statements only; 2 = uses Chemistry vocabulary with some evidence; 3 = uses Chemistry vocabulary, cites lesson evidence, and explains model change). Teacher records rubric scores as a summative indicator for the sub-strand.

	teacher asks: 'What do you see on these walls that was not in your head ten lessons ago?'	these walls came from questions you asked, evidence you gathered, and Chemistry you learned. This is what scientists do. This is what you are.' Allow 30 seconds of silent appreciation before dismissing.		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. FINAL EXPLANATION QUALITY — Did learners' individual Final Explanation paragraphs show genuine conceptual growth from Lesson 1, or did some merely repeat memorised definitions? What does this tell you about which concepts need reinforcement at the start of Sub-Strand 1.2?
2. POSTER PROJECT — Did the posters demonstrate that learners could transfer Chemistry knowledge to a real-world public health context, or did some groups default to generic health slogans without Chemistry evidence? How will you build stronger evidence-transfer skills in future units?
3. DQB COMPLETION — Which questions on the DQB were most contested (hard to categorise as answered)? These reveal the sub-strand's most challenging conceptual territory — note them for your planning of Sub-Strand 1.2 (The Atom), where many 'why does matter behave this way?' questions will be revisited.
4. PHENOMENON CONNECTION — Did learners authentically connect the chapati phenomenon to drug and substance use, or did this feel like a forced link? If the connection felt thin, consider how you will design the anchoring phenomenon for Sub-Strand 1.2 to build tighter coherence across the storyline.
5. INCLUSION & VOICE — Over the ten lessons, whose voices dominated the DQB, the model-building, and the Final Explanation? Who was silent or passive? Use Lesson 10's gallery walk and written work to identify learners who may have disengaged — reach out individually before Sub-Strand 1.2 begins.
6. PCI INTEGRATION — Did the drug and substance use discussion feel genuine and learner-owned, or was it teacher-directed? Strong PCI integration in a CBE classroom is learner-initiated — note strategies for the next unit that give learners more agency in surfacing these issues.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed (Lesson 1): raw chapati dough placed on a hot jiko transforms irreversibly into a firm, golden, aromatic disc. The original soft dough cannot be recovered. This simple, familiar Kenyan kitchen event launched ten lessons of inquiry.
What did I learn?	Learners learned: Chemistry is the scientific study of matter and its changes — physical and chemical. The branches of Chemistry (organic, inorganic, physical, biochemistry, analytical, nuclear, and others) explain different aspects of how matter behaves. Chemistry underpins agriculture in the Rift Valley, pharmaceutical production, food processing (githeri, ugali, mandazi), manufacturing, medicine, and sport. Drugs are chemical substances that interact with body matter at a molecular level; prescription, dosage, and responsible use are Chemistry-informed concepts that protect health. Matter exists as solid, liquid, or gas, with particle arrangement determining properties. Understanding Chemistry empowers learners to make safer, smarter choices in their lives.
How does this explain the phenomenon?	The driving question is answered: Chemistry explains what happens to the matter around us every day by studying the composition, properties, and changes of matter — from the irreversible chemical transformation of chapati dough on a jiko, to the way pharmaceutical drugs interact with body cells. Understanding Chemistry helps us make safer, smarter choices because it gives us evidence-based tools to evaluate food labels, understand drug prescriptions and dosages, recognise the risks of substance misuse, and appreciate the role of Chemistry-based industries in our daily Kenyan lives. The chapati cannot become dough again because a chemical change has occurred — bonds have broken and new substances have formed. Chemistry is the

	science that explains this, and every branch of Chemistry adds another layer of understanding to the matter around us.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.